

Fluid Mechanics and Turbo-Machinery (MEP221s)

Lecture (2)

Pressure and Fluid Statics (Hydrostatics)

Hydrostatic Force and Fluid in Rigid Body Rotation

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Course Contents

Fluid Mechanics and Turbo-Machinery (MEP221)

1. Properties of Fluids, Density, Pressure, Viscosity, Viscous Boundary Layers
2. Fluid Statics and Pressure Measurement,
3. Forces on Submerged Bodies,
4. Transport Theorem on a Control Volume, Continuum Hypothesis, Streamlines, Velocity and Acceleration,
5. Basic Laws: Continuity Equation, Classification of Flow Fields: Pipe Flow, Jet Flow, Wake Flow, Boundary Layer Flow
6. Similitude and Dimensional Analysis, Lagrangian and Eulerian Coordinates,
7. Compressible Flow, Stagnation Properties, Mach Number and Sonic Velocity, Equations of Gas Dynamics, Flow through Nozzles, Shock Waves,
8. Classification of Turbomachines, Operation of Pumps, Series and Parallel Operation, Selection of Pumps.
9. Flow around Immersed Bodies, Drag and Lift Forces,
10. Flow in Closed Conduits, Bernoulli's Equation, Major and Minor Losses in Pipes, Laminar and Turbulent Flows,
11. Differential Analysis of Fluid Flow: Navier-Stokes Equation,

References, Books and Scientific Journals

References and Books

1. Yunus A. Cengel & John Cimbala, "Fluid Mechanics Fundamentals and Applications" Third Edition in SI units, McGraw Hill, 2014.
2. F. M. White, "Fluid Mechanics," 7th Ed., McGraw-Hill, Seventh international edition, 2012.
3. R. Munson, D. F. Young and T. H. Okishi, "Fundamentals of Fluid Mechanics," Sixth Edition, John Wiley & Sons, Inc , ISBN 0-471-24011-2, 2009.
4. John F. Douglas, "Fluid Mechanics," 5th Ed. Pearson, Printice Hall, 2005.
5. Streeter, V.L., Wylie, E.B. and Bedford, K.W., "Fluid Mechanics "Ninth edition, McGraw Hill, 2004.
6. Seppo A. Korpela, "Principles of Turbomachinery", **2011** John Wiley & Sons, Inc.
7. Rama. S.R. Gorla, Aijaz A. Khan "Turbomachinery Design and Theory," Marcel Dekker, 2003.
8. Thomas H. McCloskey, "Handbook of Turbomachinery," Hh. 8, Steam Turbines, Marcel Dekker, 2003.

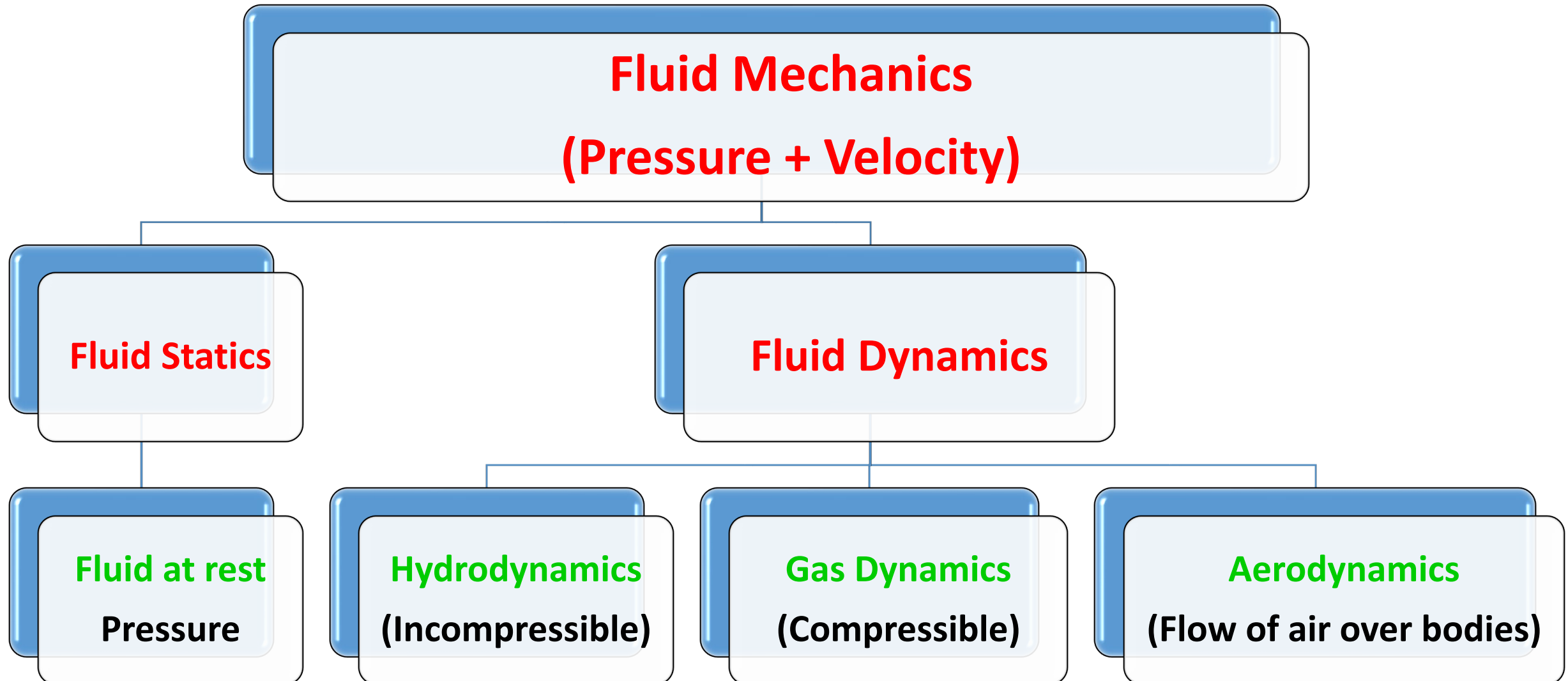
Scientific Journals

- JOURNAL OF FLUID MECHANICS
- PHYSICS OF FLUIDS
- ASME Journal of Fluids Engineering
- Engineering Applications of Computational Fluid Mechanics
- COMPUTERS & FLUIDS
- JOURNAL OF TURBULENCE
- Journal of Turbomachinery – Transaction of ASME
- International Journal of Turbomachinery, Propulsion and Power

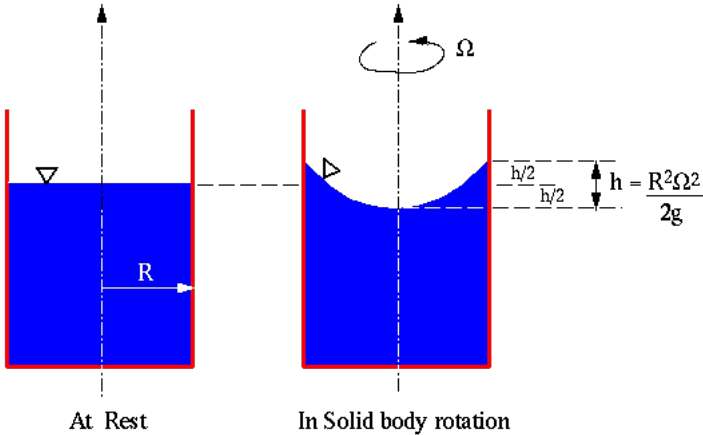
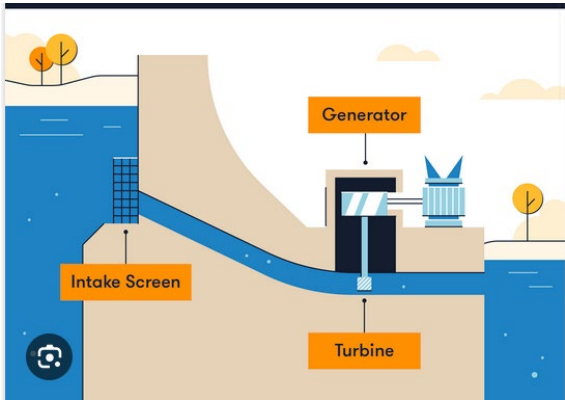
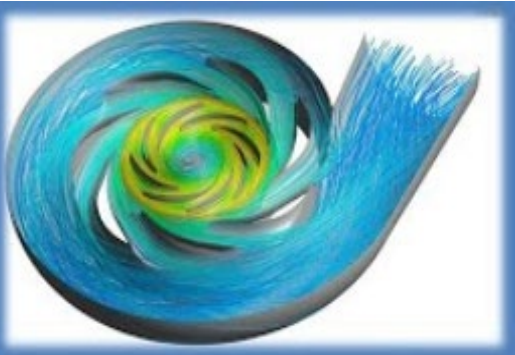
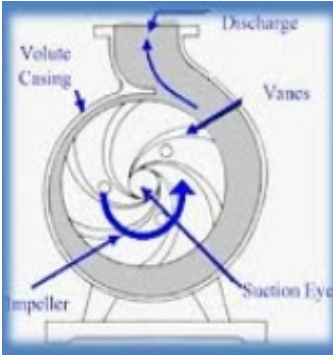
Outlines

Fluid Statics (Hydrostatics)

1. Absolute, Gage and Vacuum Pressure.
2. The Basic Equation of Fluid Statics.
3. Pressure Variation in a Static Fluid.
4. The Standard Atmosphere.
5. Manometers.
6. Barometers.
7. Variation of pressure in gases.
8. Hydrostatic Force on Submerged Surfaces.



Practical Applications of Fluid Statics (Hydrostatics)



Hydrostatic force on Submarine and ships

Hydrostatic force on dams and sluice gates

Rotating Fluids (Rigid bod rotation)
Forced vortex motion
Example: Centrifugal pump

Practical Application - Hydrostatic Force on Submerged Surfaces

When a surface is submerged in a fluid, forces develop on the surface due to the fluid.

The determination of these forces is important in the design of storage tanks, ships, dams, and other hydraulic structures

For fluids at rest we know that the force must be perpendicular to the surface since there are no shearing stresses present

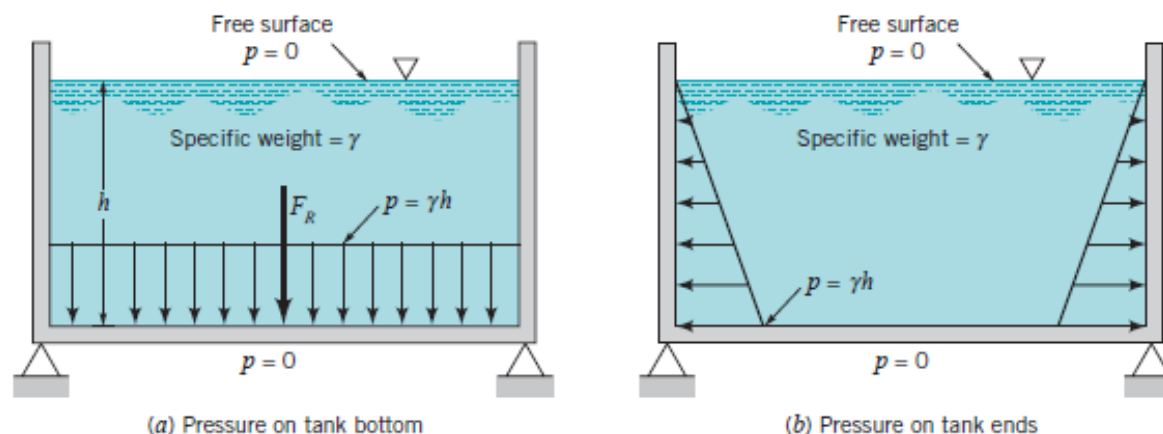


FIGURE 2.16 (a) Pressure distribution and resultant hydrostatic force on the bottom of an open tank. (b) Pressure distribution on the ends of an open tank.

Fluid Statics (Hydrostatics)

When fluids are in statics → No shear stress exists

$$\tau = \mu \frac{dv}{dy} \rightarrow v = 0 \text{ [m/s]} \rightarrow \tau = 0$$

Shear stresses due to viscosity are only generated when there is relative motion between elements of the fluid

The only forces that develop on the surfaces of the particles will be due to the pressure.

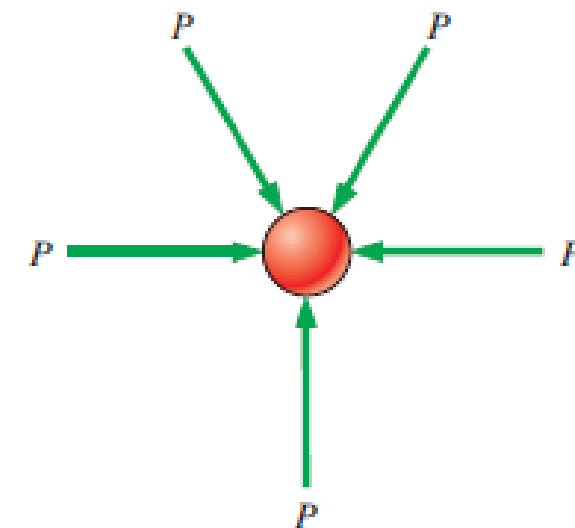
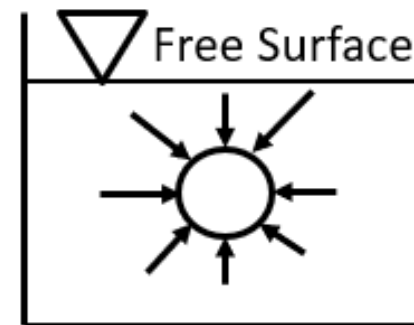
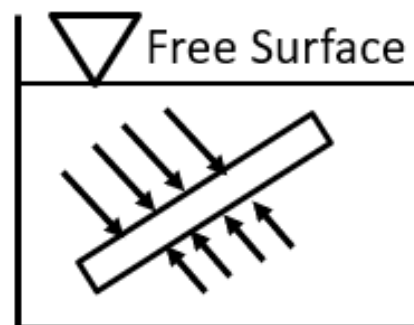
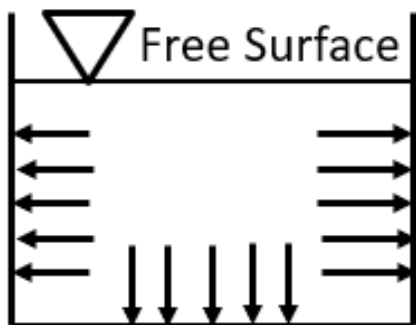


FIGURE 3-4

Pressure is a *scalar* quantity, not a vector; the pressure at a point in a fluid is the same in all directions.

Definition of Pressure

Pressure is defined as a normal force exerted by a fluid per unit area.

$$p = \frac{F_{normal}}{A} \text{ [Pa]}$$

SI units: Newton/square meters (N/m²) = Pascal (Pa).

1 bar = 10⁵ Pa = 0.1 MPa = 100 kPa

There are **no shearing stresses** present in a fluid at **rest**.

The **only forces** that develop on the surfaces of the particles will be due to the **pressure**.

Fluids in contact with surfaces exert a force on the surfaces. The force is mainly due to the **specific weight** of the fluid in the case of **liquids**.

In the case of **gases**, **molecular activity** is the main cause of force exerted on the surfaces of the containers. Gas column will also exert a force on the base, but this is usually small in magnitude.

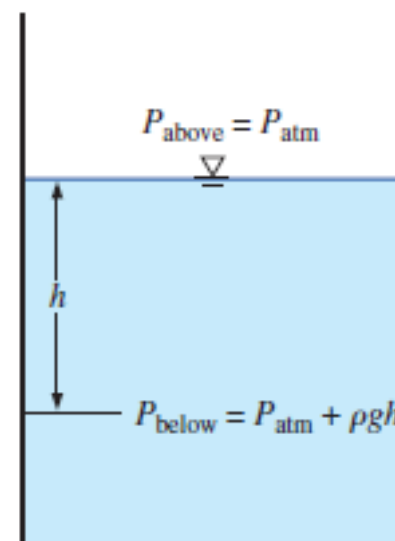
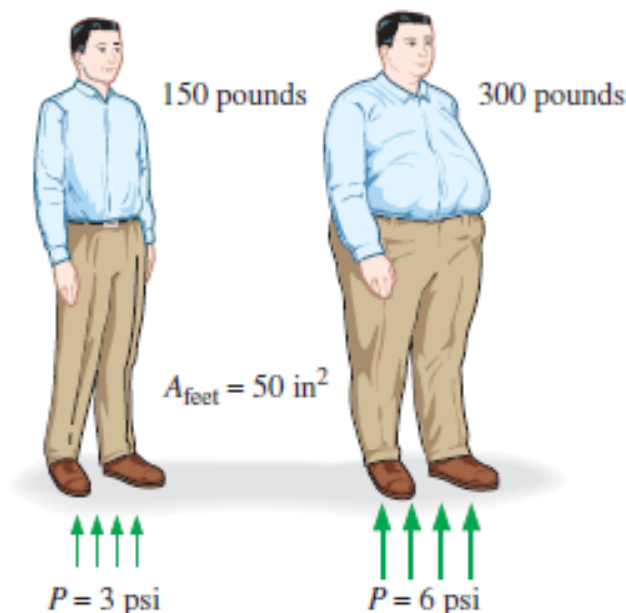


FIGURE 3–9

Pressure in a liquid at rest increases linearly with distance from the free surface.



$$P = \sigma_n = \frac{W}{A_{feet}} = \frac{150 \text{ lbf}}{50 \text{ in}^2} = 3 \text{ psi}$$

FIGURE 3–1

The normal stress (or “pressure”) on the feet of a chubby person is much greater than on the feet of a slim person.

Absolute, Gage and Vacuum Pressures

Absolute pressure:

The actual pressure at a given position is called the **absolute pressure**, and it is measured relative to absolute vacuum (i.e., absolute zero pressure).

Gage pressure:

Most pressure-measuring devices, however, are calibrated to read zero in the atmosphere and so they indicate the difference between the absolute pressure and the local atmospheric pressure. This difference is called the **gage pressure**

$$p_{\text{gage}} = p_{\text{abs}} - p_{\text{atm}}$$

$$p_{\text{abs}} > p_{\text{atm}} \rightarrow p_{\text{gage}} = +ve \text{ (gage)}$$

Vacuum pressure:

Pressure below atmospheric pressure are called **vacuum pressure** and is measured by vacuum gage that indicates the difference between the atmospheric pressure and the absolute pressure.

$$p_{\text{vac}} = p_{\text{atm}} - p_{\text{abs}}$$

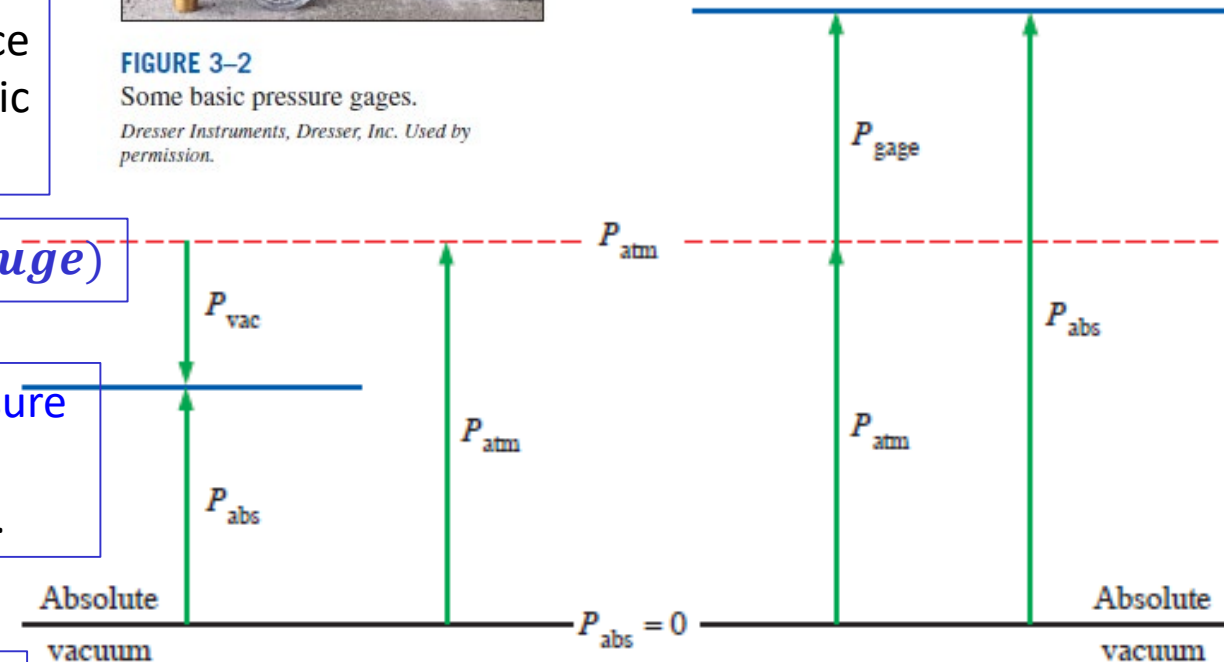
$$p_{\text{abs}} < p_{\text{atm}} \rightarrow p_{\text{gage}} = -ve \text{ (vacuum) or (suction)}$$



FIGURE 3-2

Some basic pressure gages.

Dresser Instruments, Dresser, Inc. Used by permission.



Absolute, gage, and vacuum pressures

Note:

Absolute, gage, and vacuum pressures are all positive quantities and are related to each other

The Basic Equation of Fluid Statics (Hydrostatics)

For a differential fluid element, the **body force** is:

$$d\vec{F}_B = \vec{g} dm = \vec{g} \rho dV$$



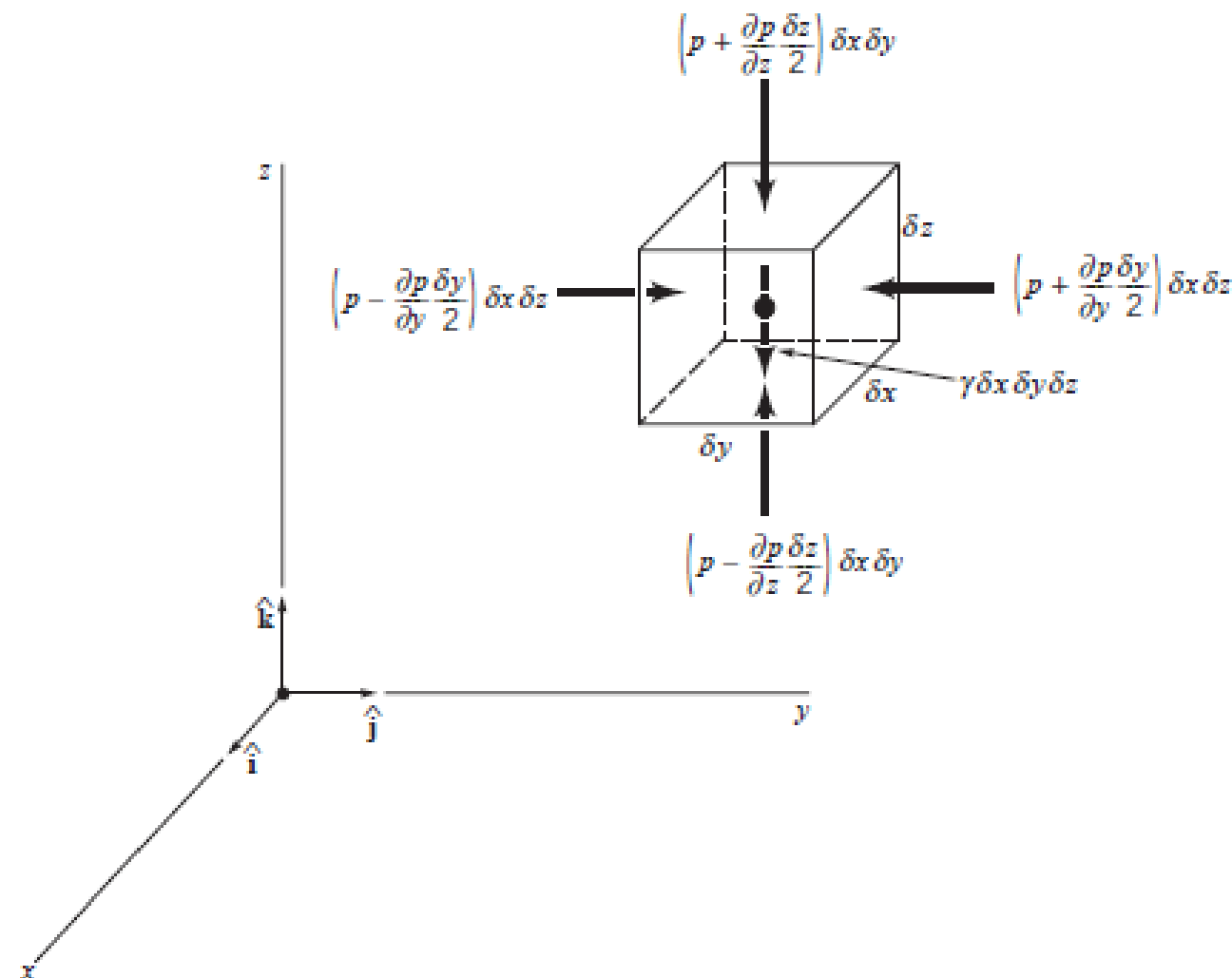
$$d\vec{F}_B = \vec{g} \rho dx dy dz$$

Use a **Taylor series expansion** of the **pressure** about **point O**, the pressure on the left and right surfaces are:



$$p_{\text{left}} = p + \left(\frac{\partial p}{\partial y} \right) \left(-\frac{dy}{2} \right) = p - \frac{\partial p}{\partial y} \frac{dy}{2}$$

$$p_{\text{right}} = p + \left(\frac{\partial p}{\partial y} \right) \left(\frac{dy}{2} \right) = p + \frac{\partial p}{\partial y} \frac{dy}{2}$$



■ **FIGURE 2.2** Surface and body forces acting on small fluid element. 11

The Basic Equation of Fluid Statics (Hydrostatics)

The **net surface force** acting on the element is:

$$\begin{aligned}
 d\vec{F}_s &= \left(p - \frac{\partial p}{\partial x} \frac{dx}{2} \right) (dy \, dz) (\hat{i}) + \left(p + \frac{\partial p}{\partial x} \frac{dx}{2} \right) (dy \, dz) (-\hat{i}) \\
 &+ \left(p - \frac{\partial p}{\partial y} \frac{dy}{2} \right) (dx \, dz) (\hat{j}) + \left(p + \frac{\partial p}{\partial y} \frac{dy}{2} \right) (dx \, dz) (-\hat{j}) \\
 &+ \left(p - \frac{\partial p}{\partial z} \frac{dz}{2} \right) (dy \, dx) (\hat{k}) + \left(p + \frac{\partial p}{\partial z} \frac{dz}{2} \right) (dy \, dx) (-\hat{k})
 \end{aligned}$$

pressure gradient

$$\vec{dF}_s = - \left(\frac{\partial p}{\partial x} \hat{i} + \frac{\partial p}{\partial y} \hat{j} + \frac{\partial p}{\partial z} \hat{k} \right) (dx \, dy \, dz)$$

The term in parentheses is called the **gradient** of the **pressure** or simply the **pressure gradient** and may be written **grad p** or ∇p . In rectangular coordinates.

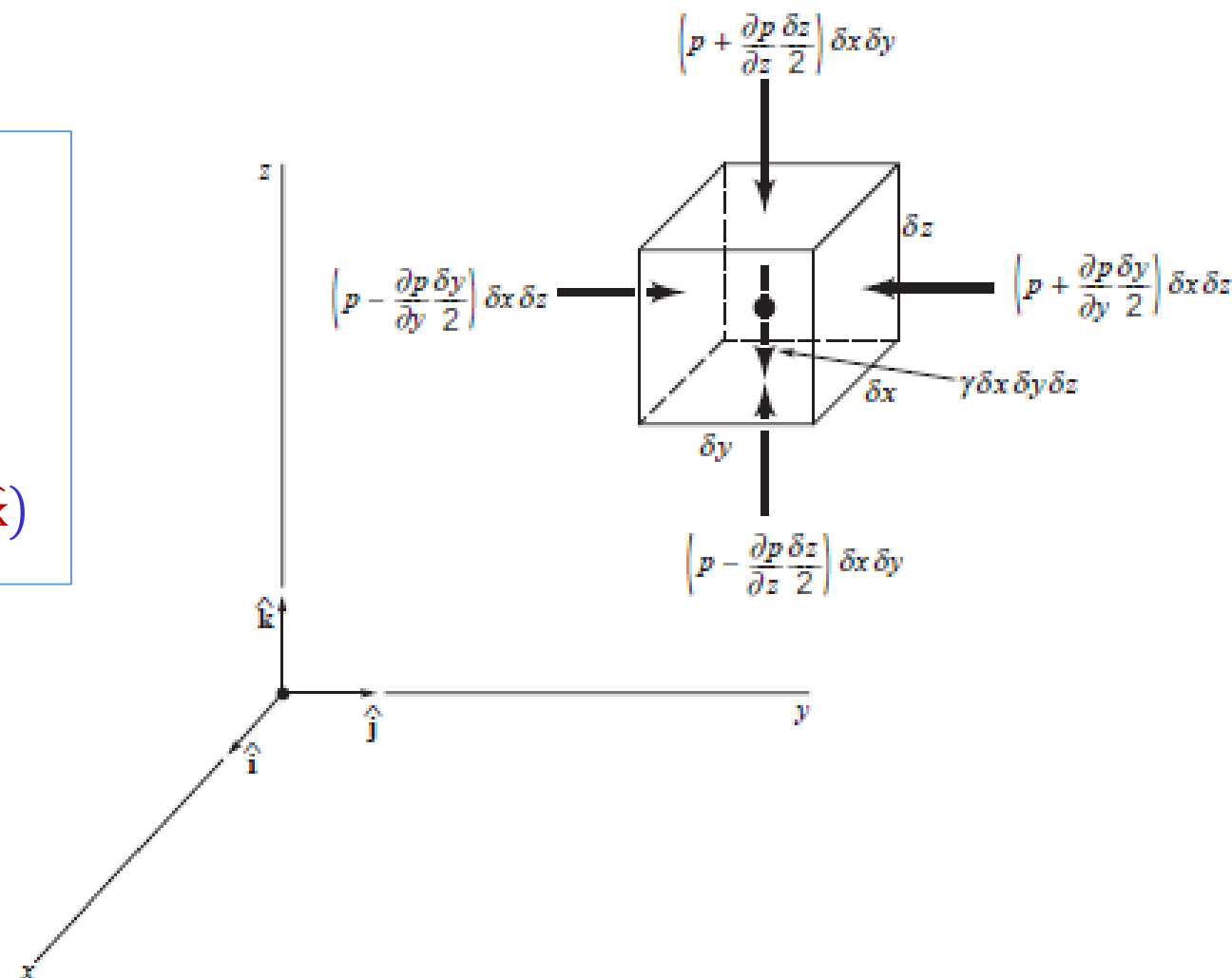


FIGURE 2.2 Surface and body forces acting on small fluid element. 12

The Basic Equation of Fluid Statics (Hydrostatics)

The pressure gradient:

$$\text{grad } p \equiv \nabla p \equiv \left(\frac{\partial p}{\partial x} \hat{i} + \frac{\partial p}{\partial y} \hat{j} + \frac{\partial p}{\partial z} \hat{k} \right) \equiv \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) p$$

$$d\vec{F}_s = -\text{grad } p(dx dy dz) = -\nabla p dx dy dz$$

Combine the **surface** and **body** forces that we have developed to obtain the **total force** acting on a fluid element:

$$d\vec{F} = d\vec{F}_s + d\vec{F}_B$$

$$d\vec{F} = (-\nabla p + \rho \vec{g}) dx dy dz = (-\nabla p + \rho \vec{g}) dV$$

Per unit volume basis:

$$\frac{d\vec{F}}{dV} = (-\nabla p + \rho \vec{g})$$

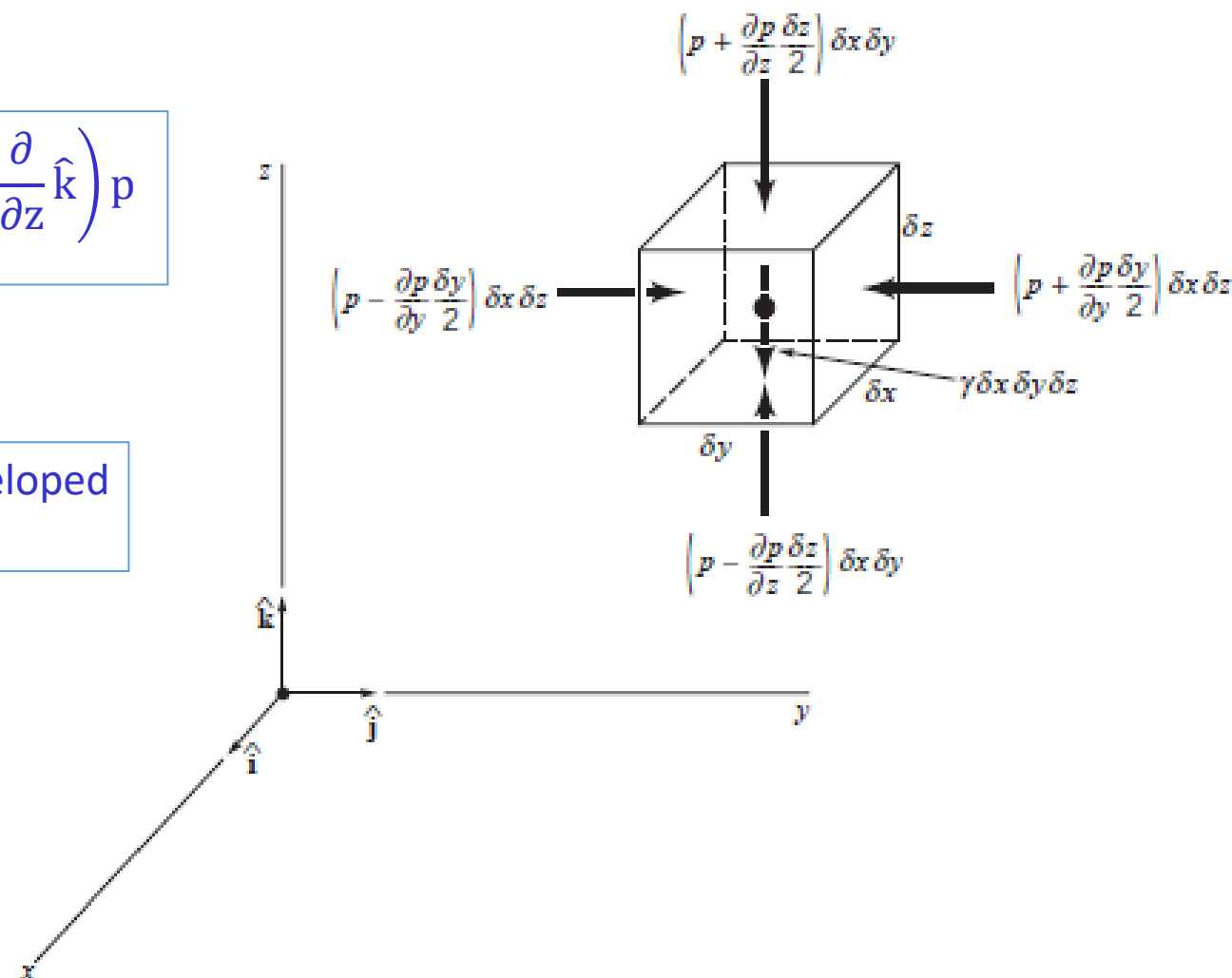


FIGURE 2.2 Surface and body forces acting on small fluid element.

The Basic Equation of Fluid Statics (Hydrostatics)

For a fluid particle, **Newton's second law** gives:

$$\vec{F} = \frac{d}{dt}(m\vec{V})$$

For a **static fluid**:

$$\vec{V} = 0$$

Thus:

$$\frac{d\vec{F}}{dV} = 0$$

$$\begin{aligned} -\frac{\partial p}{\partial x} + \rho g_x &= 0 \rightarrow x \text{ direction} \\ -\frac{\partial p}{\partial y} + \rho g_y &= 0 \rightarrow y \text{ direction} \\ -\frac{\partial p}{\partial z} + \rho g_z &= 0 \rightarrow z \text{ direction} \end{aligned}$$

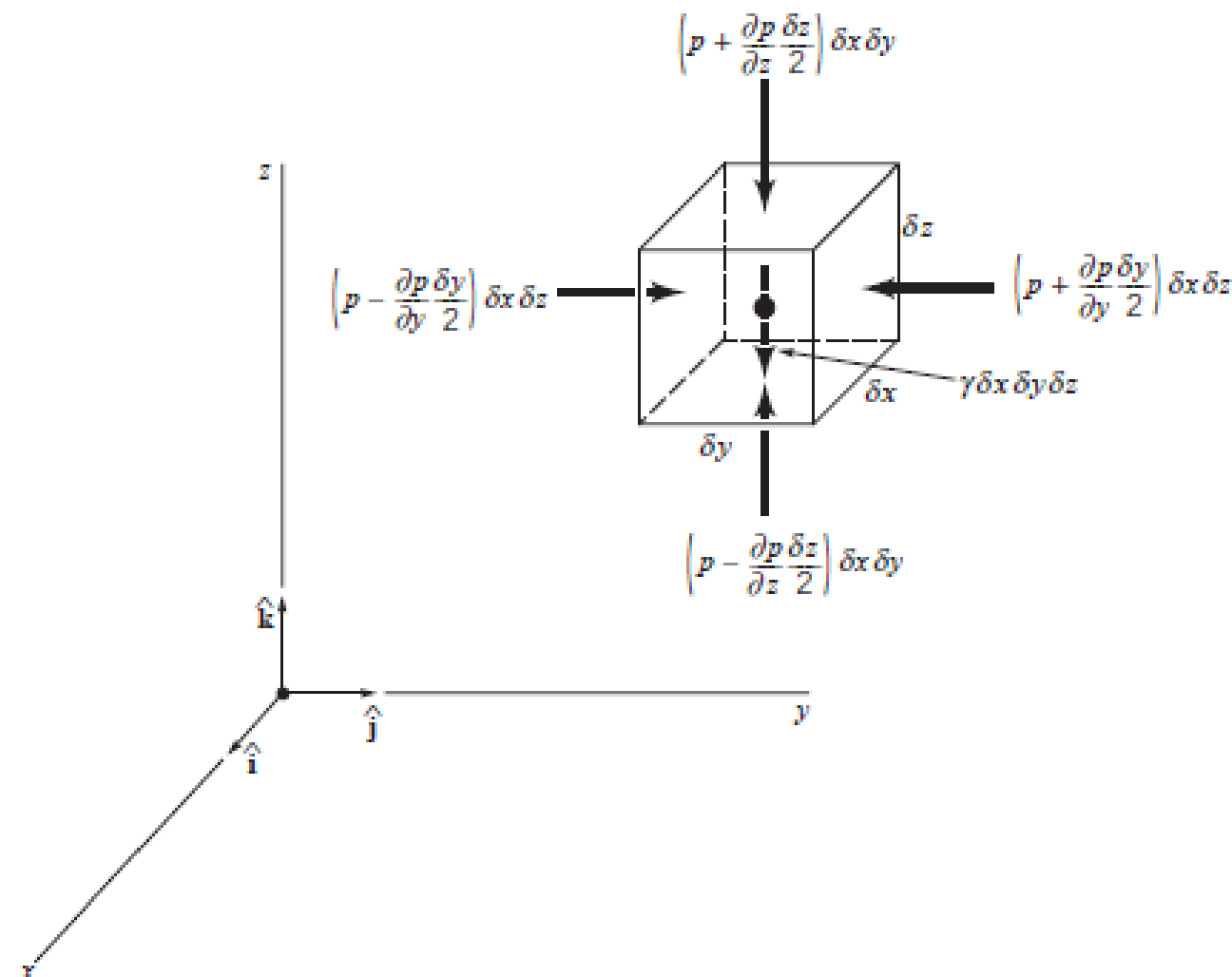


FIGURE 2.2 Surface and body forces acting on small fluid element. 14

The Basic Equation of Fluid Statics (Hydrostatics)

But

$$g_z = -g$$

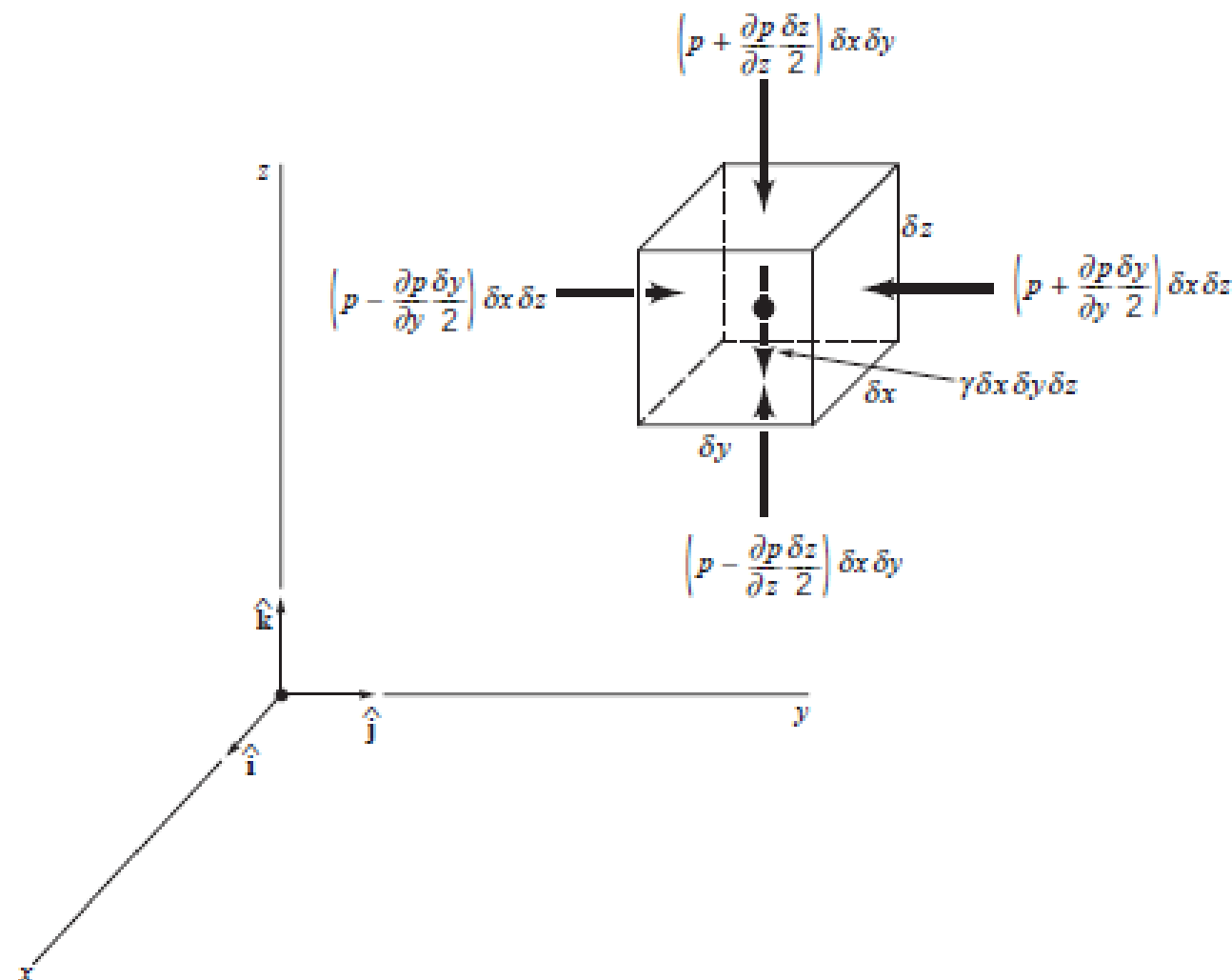
$$g_x = 0$$

$$g_y = 0$$

$$\Rightarrow \frac{\partial p}{\partial z} = -\rho g = -\gamma$$

Conditions of use:

- (1) Static fluid.
- (2) Gravity is the only body force.
- (3) The z axis is vertical and upward



■ **FIGURE 2.2** Surface and body forces acting on small fluid element. 15

Pressure Variation in a Static Fluid and Pascal's Law

(Pressure gradient of fluids at rest)

$$\frac{dp}{dz} = -\rho g$$

Pascal's law:

"pressure has the same value at the horizontal plan for the same fluid"

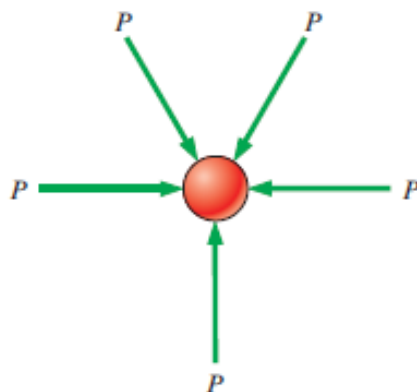


FIGURE 3-4

Pressure is a *scalar* quantity, not a vector; the pressure at a point in a fluid is the same in all directions.

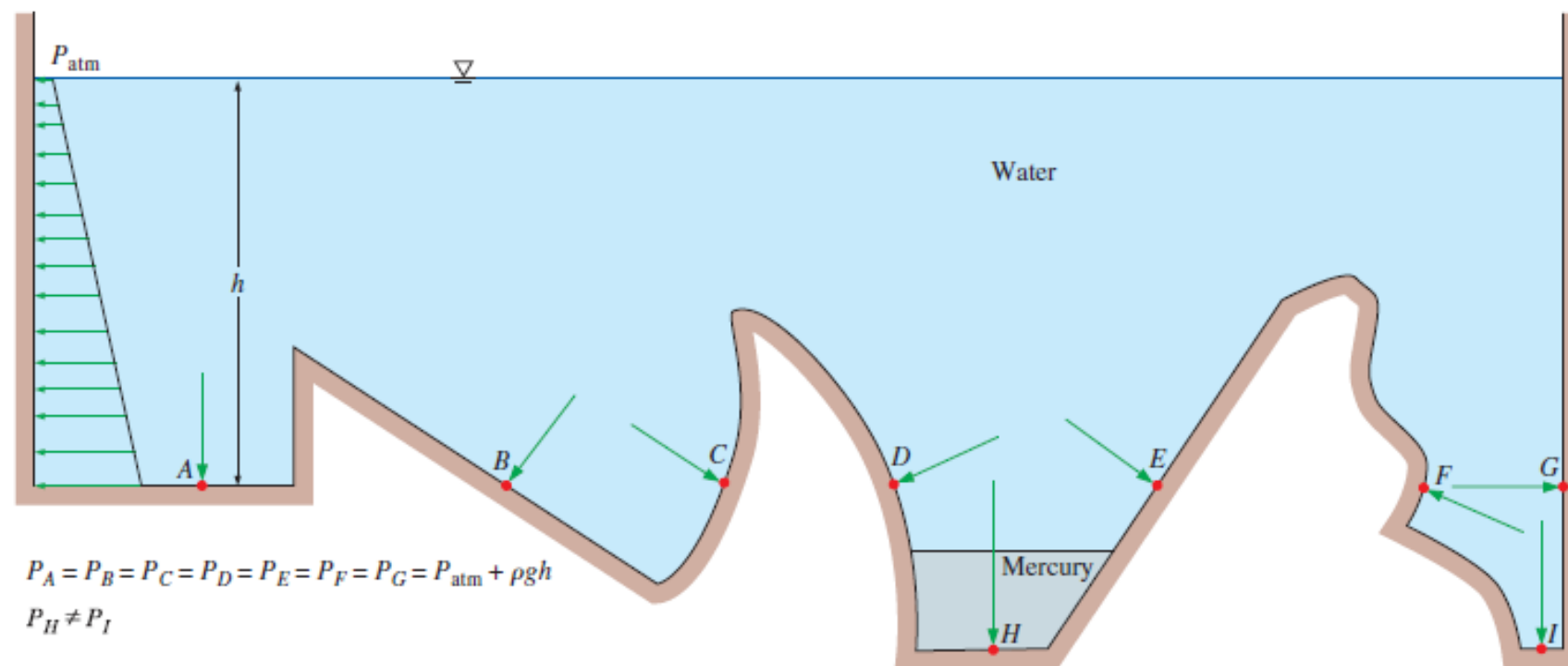


FIGURE 3-10

Under hydrostatic conditions, the pressure is the same at all points on a horizontal plane in a given fluid regardless of geometry, provided that the points are interconnected by the same fluid.

Pressure and Pressure Head

From the shown figure, pressure exerted by fluid at an element area (dA):

$$\rightarrow p = \frac{dW}{dA} = \frac{\gamma h dA}{dA} = \gamma h \text{ [Pa]}$$

h : is pressure head [m of fluid]
 γ : is specific weight of fluid [N/m^3]

$$\rightarrow h = \frac{p}{\gamma} \text{ [m of fluid]}$$

For constant pressure, pressure can be expressed in different fluid as follows,

$$\rightarrow p = \gamma_1 h_1 = \gamma_2 h_2 = \gamma_3 h_3 = \dots \text{ [Pa]}$$

Example:

$$p_{atm} = 101325 \text{ [Pa]} = 10.3 \text{ [cm Water]} = 76 \text{ [cm Mercury]}$$

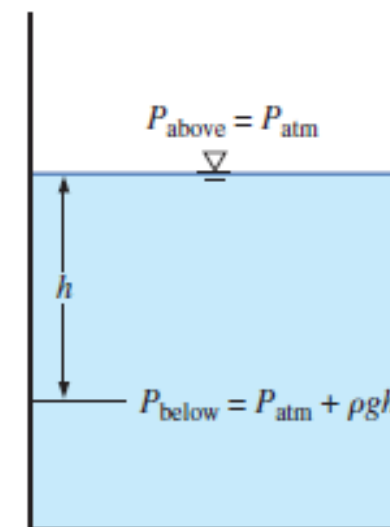
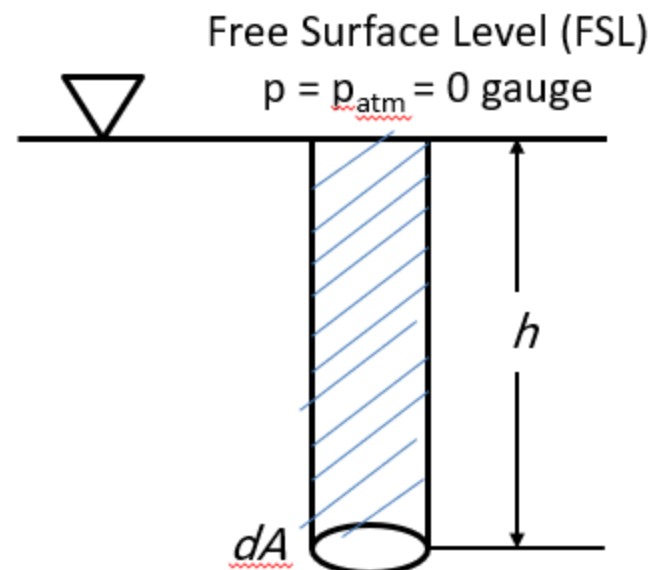


FIGURE 3–9

Pressure in a liquid at rest increases linearly with distance from the free surface.

Temperature and Pressure Distribution in the Standard Atmosphere

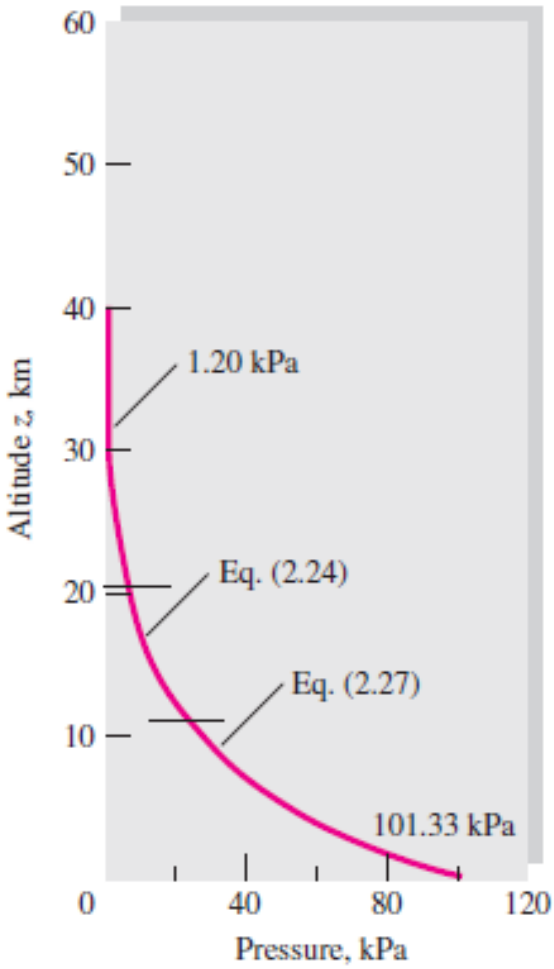
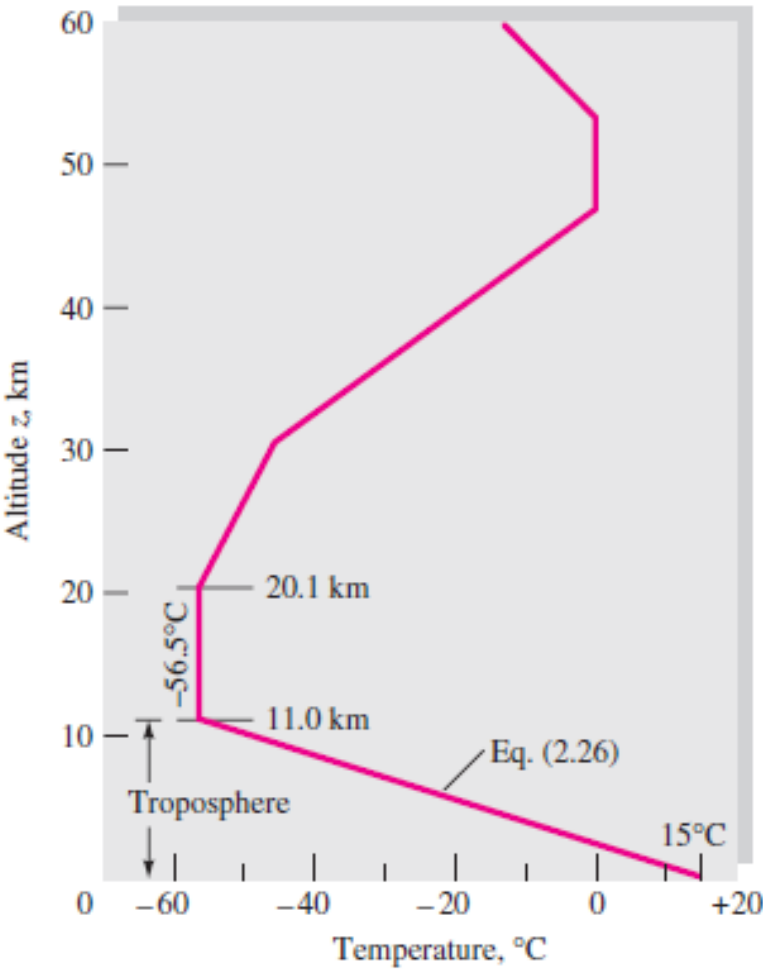
The standard atmosphere is an idealized representation of mean conditions in the earth’s atmosphere.

Properties of U.S. Standard Atmosphere at Sea Level^a

Property	SI Units	BG Units
Temperature, <i>T</i>	288.15 K (15 °C)	518.67 °R (59.00 °F)
Pressure, <i>p</i>	101.33 kPa (abs)	2116.2 lb/ft ² (abs) [14.696 lb/in. ² (abs)]
Density, <i>ρ</i>	1.225 kg/m ³	0.002377 slugs/ft ³
Specific weight, <i>γ</i>	12.014 N/m ³	0.07647 lb/ft ³
Viscosity, <i>μ</i>	1.789 × 10 ⁻⁵ N · s/m ²	3.737 × 10 ⁻⁷ lb · s/ft ²

^aAcceleration of gravity at sea level = 9.807 m/s² = 32.174 ft/s².

As is shown in this figure the temperature decreases with altitude in the region nearest the earth’s surface (troposphere), then becomes essentially constant in the next layer (stratosphere), and subsequently starts to increase in the next layer

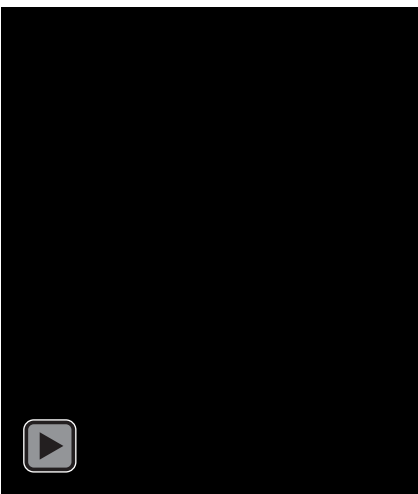
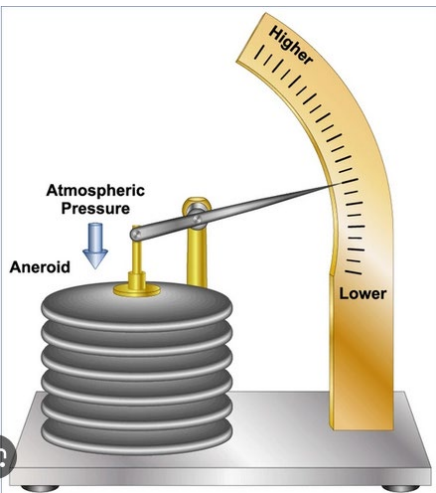
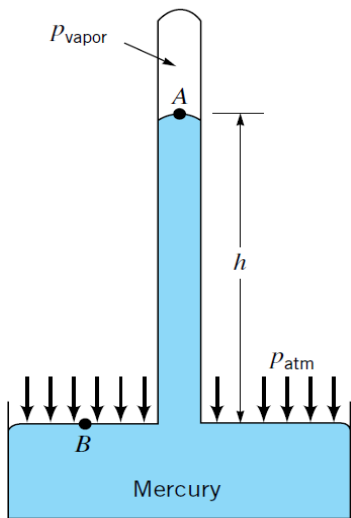


Measurement of Pressure

Measurement of Atmospheric Pressure

Mercury Barometer (Torricelli Barometer, 1644)

Mechanically (by the Aneroid Apparatus)



Measurement of Fluid Pressure

Mechanically (by the Bourdon Gauge-Bourdon Tube)

Manometers (Manometry)

Piezometer tube

U-Tube Manometer

Differential Manometer

Inclined Manometer

Inverted Differential Manometer



Manometers are devices that employ liquids to measure pressure.

1. Measurement of Atmospheric Pressure

1.1. Mercury Barometer (Torricelli Barometer, 1644)

Atmospheric pressure may be obtained from a barometer, in which the height of a mercury column is measured.

Standard atmospheric (at sea level and 20°C) pressure can be expressed as:

$$p_{atm} = \rho_{\text{Mercury}} g h_{\text{Mercury}} + p_{\text{vapor Mercury}}$$

$$p_{\text{vapor Mercury}} \approx 0$$

$$\Rightarrow p_{atm} = \rho_{Hg} \cdot g \cdot h_{Hg}$$

$$p_{atm} = 101325 \text{ Pa (abs)} = 0 \text{ Pa (gauge)}$$

$$p_{atm} = 760 \text{ mm Hg (abs)} = 13.33 \text{ m water (abs)}$$

The weight of the air column of height (H):

$$W = mg = \rho_{\text{air}} Vg = \rho_{\text{air}} AHg$$

Therefore, the atmospheric pressure is:

$$p_{atm} = \frac{W}{A} = \rho_{\text{air}} gH$$

Note that ρ_{air} is function of H

This device attributed to **Evangelista Torricelli** in about 1644

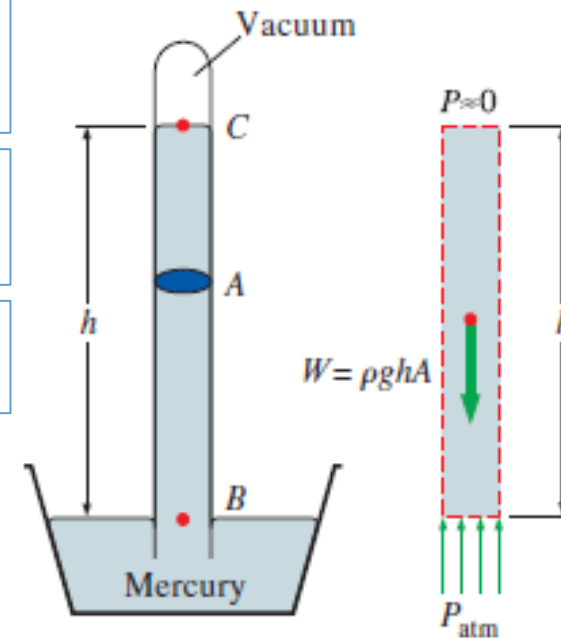


FIGURE 3-12

The basic barometer.

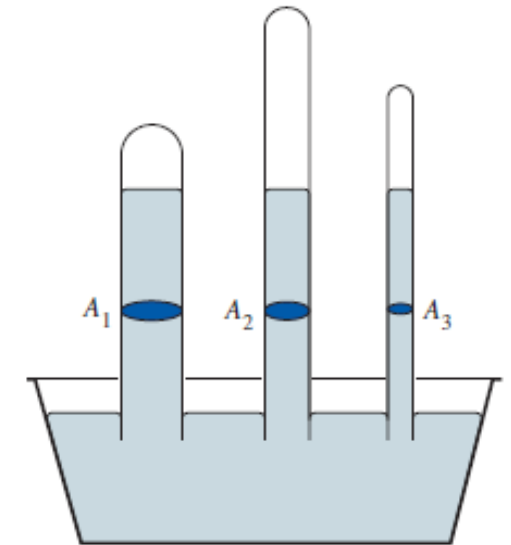
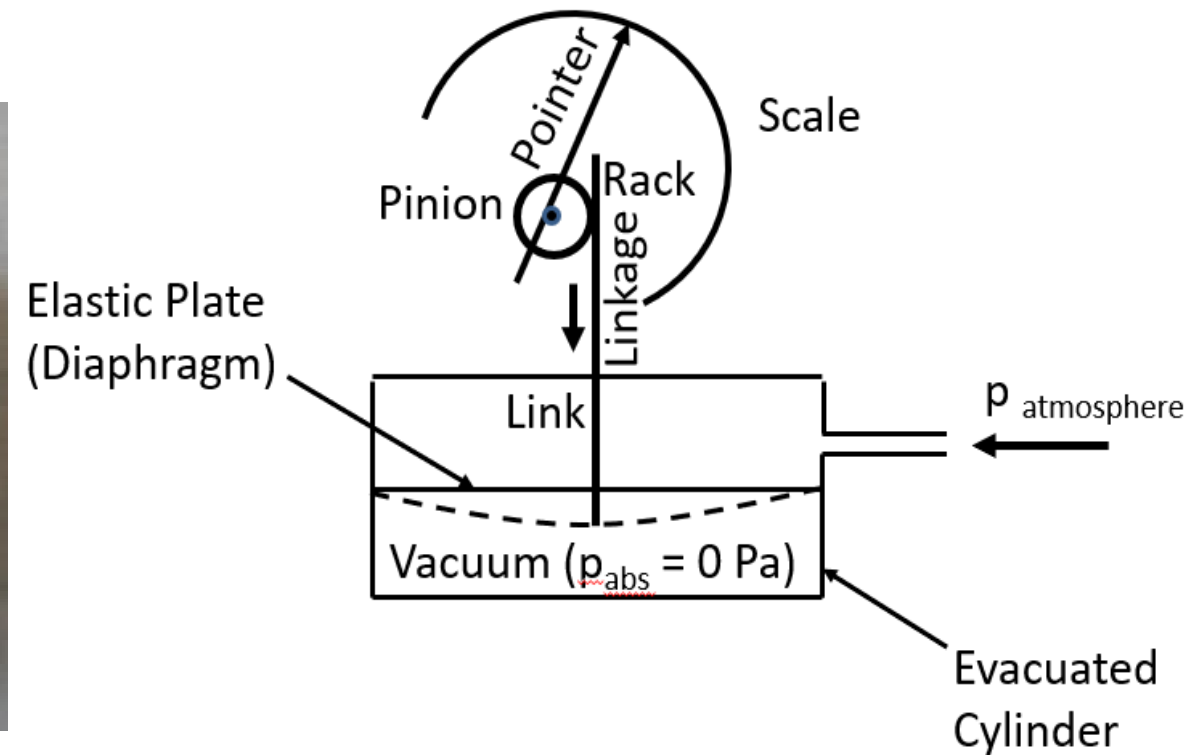
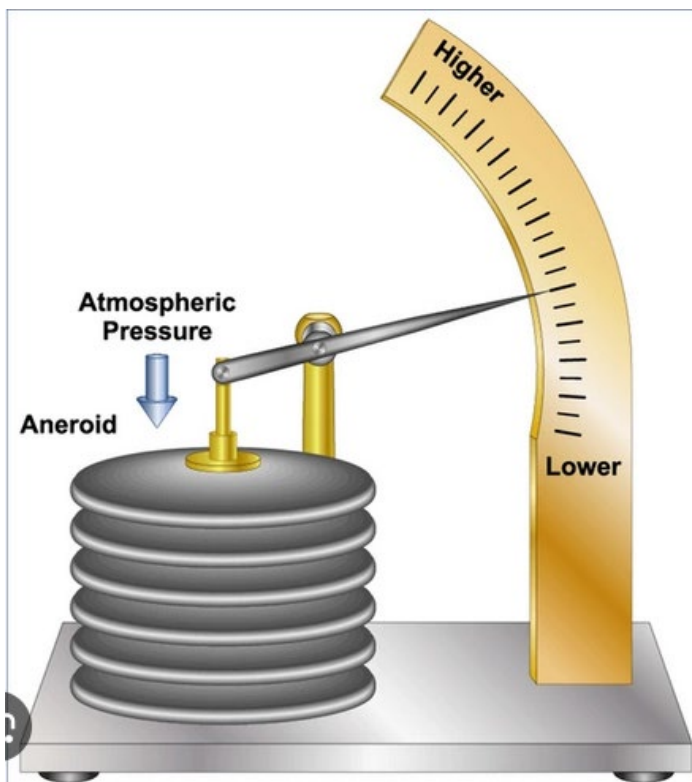


FIGURE 3-13

The length or the cross-sectional area of the tube has no effect on the height of the fluid column of a barometer, provided that the tube diameter is large enough to avoid surface tension (capillary) effects.

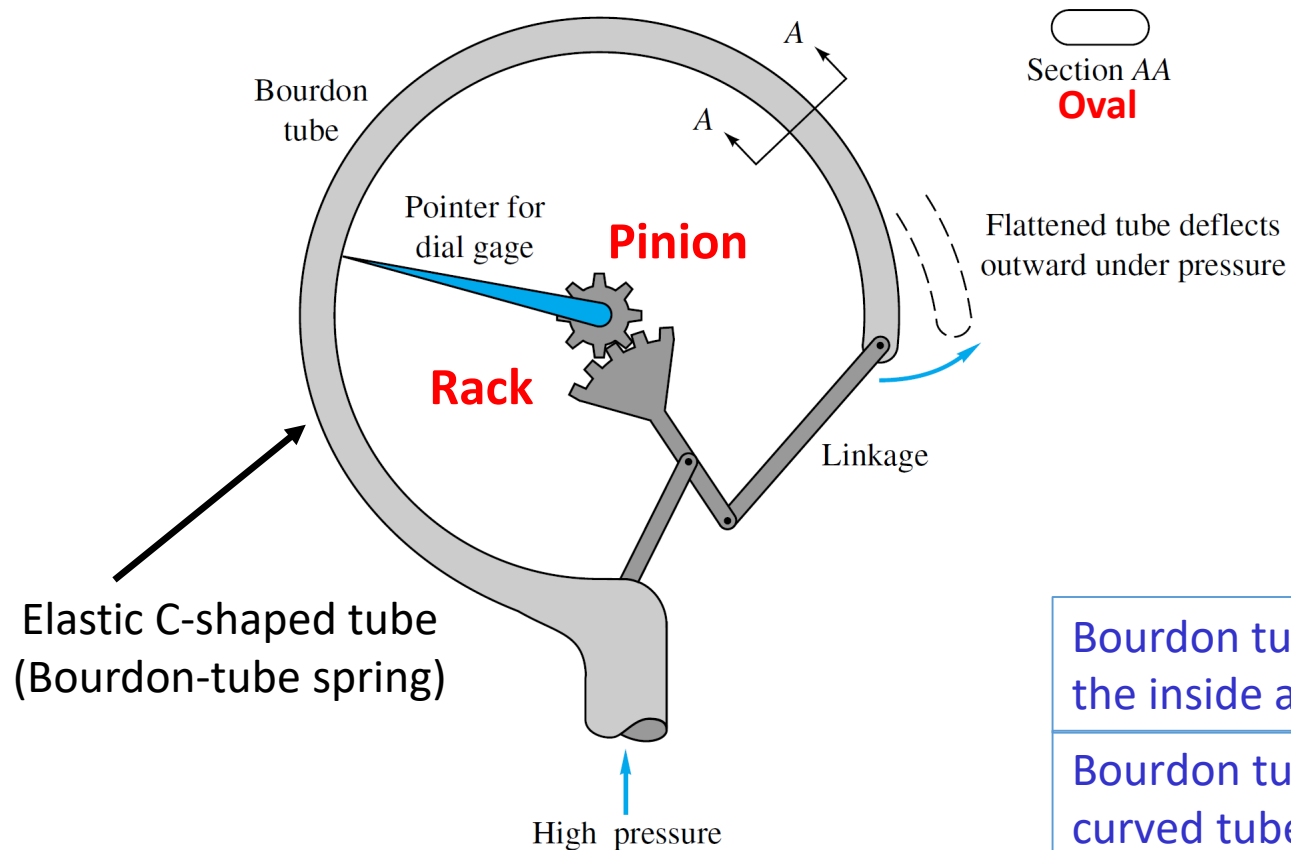
1. Measurement of Atmospheric Pressure

1.1. Mechanically by using the Aneroid Apparatus



2. Measurement of Fluid Pressure

2.1. Mechanically (by the Bourdon Gauge- Bourdon Tube)



$$\text{Reading} = p_{\text{inside}} - p_{\text{outside}}$$

Bourdon tube measures the pressure difference between the inside and the outside of the tube

Bourdon tube pressure gage uses a hollow, elastic, and curved tube to measure pressure.

A zero reading on the gage indicates that the measured pressure is equal to the local atmospheric pressure

2.2 Manometers (Manometry)

2.2.1. Piezometer Tube

Piezometer Tube Vertical shot tube to measure small positive pressure

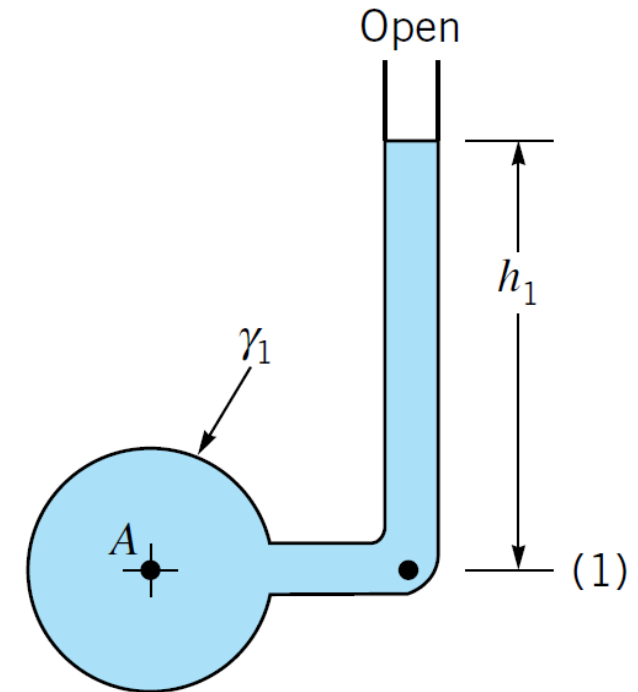
$$p_{A(gauge)} = \gamma_1 h_1 [\text{Pa}]$$

$$p_{A(absolute)} = \gamma_1 h_1 + p_{atm} [\text{Pa}]$$

Where h_1 is pressure head [m of fluid]

Disadvantages:

1. It is only suitable if the pressure in the container is greater than atmospheric pressure (otherwise air would be sucked into the system)
2. The pressure to be measured must be relatively small so the required height of the column is reasonable.
3. The fluid in the container in which the pressure is to be measured must be a liquid rather than a gas.



Piezometer Tube

2.2 Manometers (Manometry)

2.2.2. U-Tube Manometer

To measure great positive or negative pressure using a manometric (gauge) fluid, i.e., mercury in the U-tube

For an incompressible fluid, $\rho = \text{constant}$. Then for constant gravity,

$$\frac{dp}{dz} = -\rho g$$

$$p_{A,abs} + \gamma_1 h_1 = \gamma_2 h_2 + p_{atm}$$



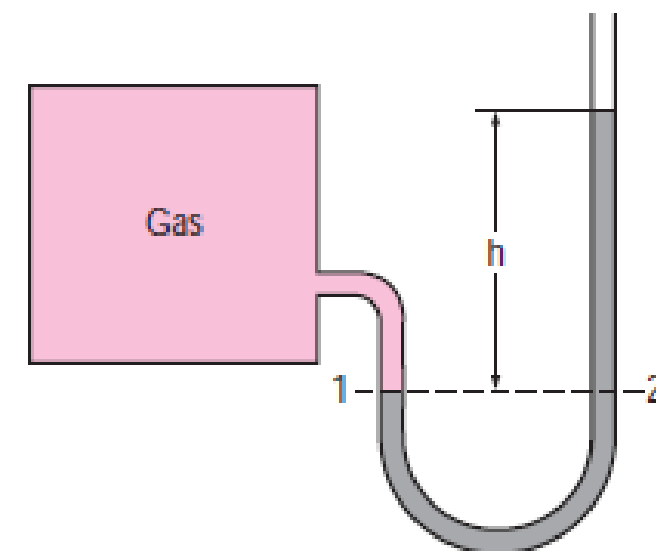
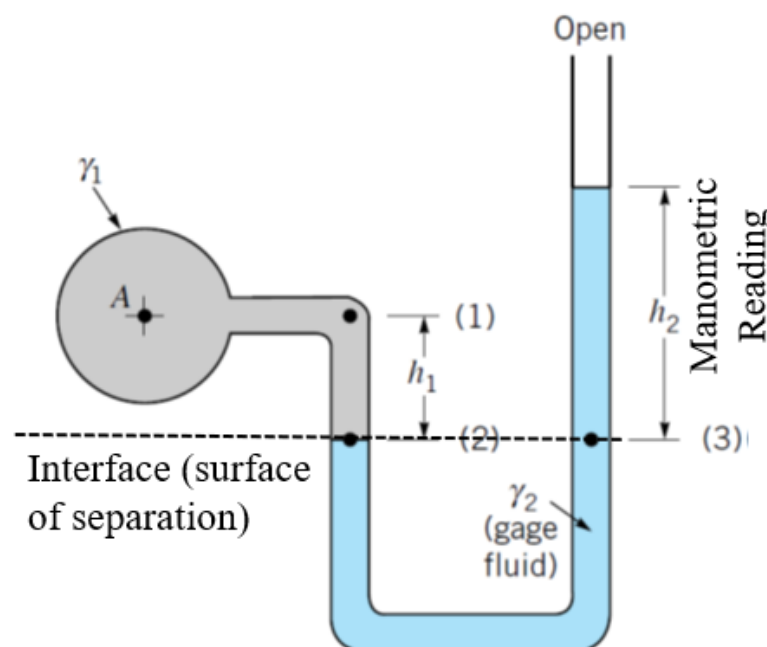
$$p_{A,gauge} + \gamma_1 h_1 = \gamma_2 h_2$$

or

$$p_{A,gauge} + \gamma_1 h_1 - \gamma_2 h_2 = 0$$

Note:

$$\gamma_2 (\text{manometric fluid}) > \gamma_1 (\text{fluid in pipe})$$



The basic manometer.

The pressure difference between two points in a static incompressible fluid can be determined by measuring the elevation difference between the two points. Devices used for this purpose are called **manometers**.

2.2 Manometers (Manometry)

2.2.3. Differential U-Tube Manometer

Differential U-Tube Manometer:

To measure pressure difference between two fluids

$$\gamma_1 < \gamma_2 \text{ (manometric fluid)} > \gamma_3$$

$$P_2 = P_3$$

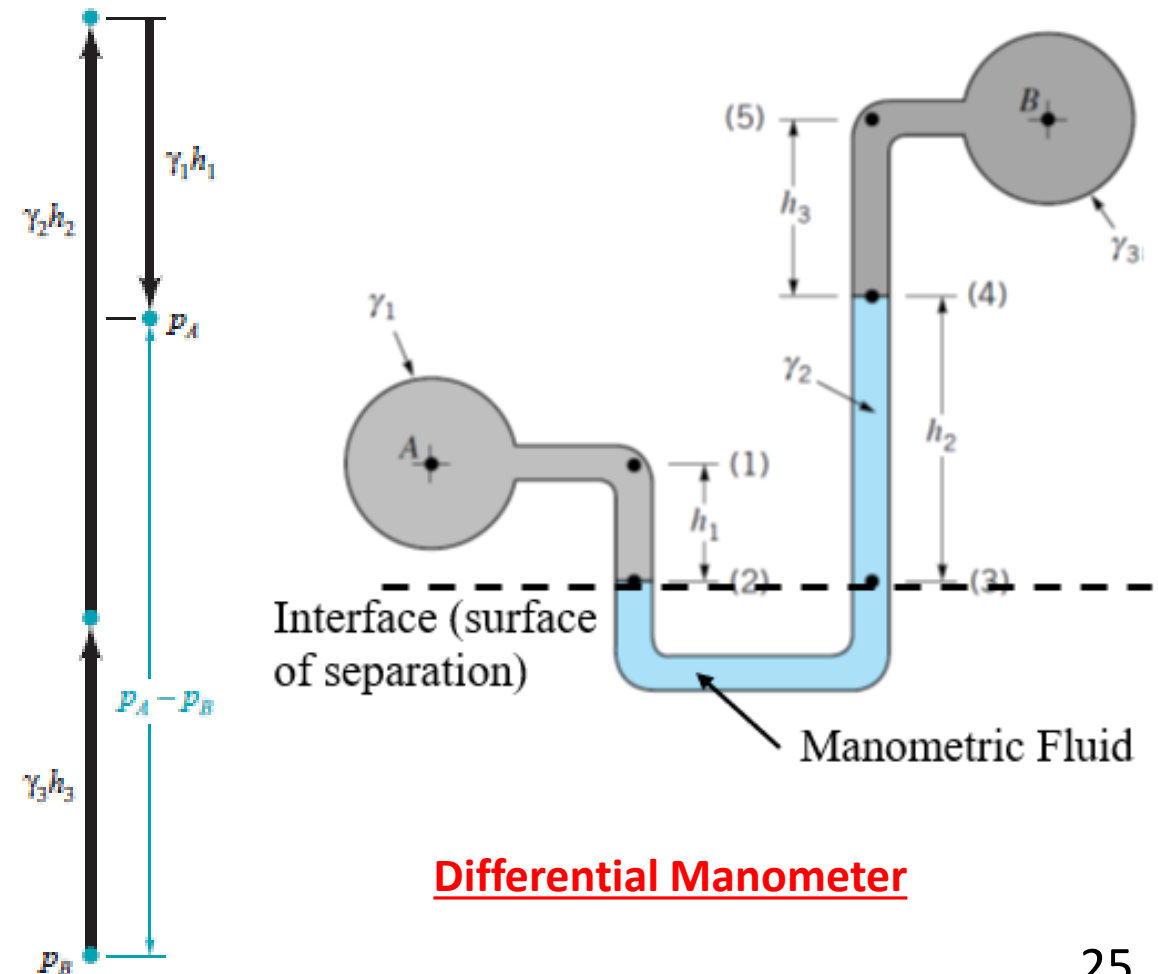
$$P_A + \gamma_1 h_1 = P_B + \gamma_2 h_2 + \gamma_3 h_3$$

$$P_A - P_B = \gamma_2 h_2 + \gamma_3 h_3 - \gamma_1 h_1$$

OR

$$P_A + \gamma_1 h_1 - \gamma_2 h_2 - \gamma_3 h_3 = P_B$$

$$P_B + \gamma_3 h_3 + \gamma_2 h_2 - \gamma_1 h_1 = P_A$$



2.2 Manometers (Manometry)

2.2.4. Inverted Differential U-Tube Manometer

For the inverted differential Manometer:

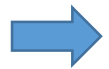
$$\gamma_1 > \gamma_2 \text{ (manometric fluid)} < \gamma_3$$



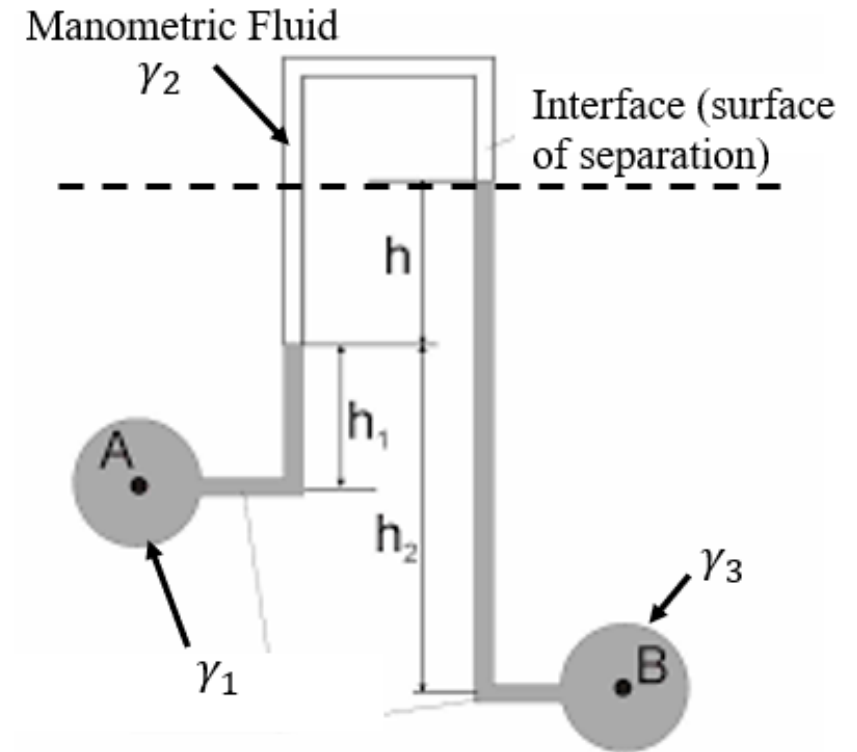
$$P_A - \gamma_1 h_1 - \gamma h = P_B - \gamma_3 (h_2 + h)$$

OR

$$P_A - \gamma_1 h_1 - \gamma h + \gamma_3 (h_2 + h) = P_B$$



$$P_A - P_B = \gamma_1 h_1 + \gamma h - \gamma_3 (h_2 + h)$$

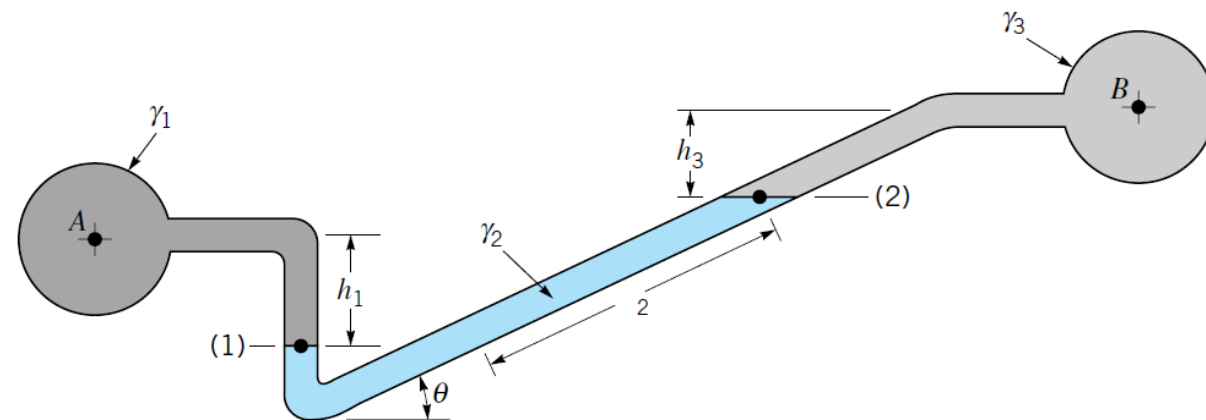


Inverted Differential Manometer

2.2 Manometers (Manometry)

2.2.5. Inclined Differential U-Tube Manometer

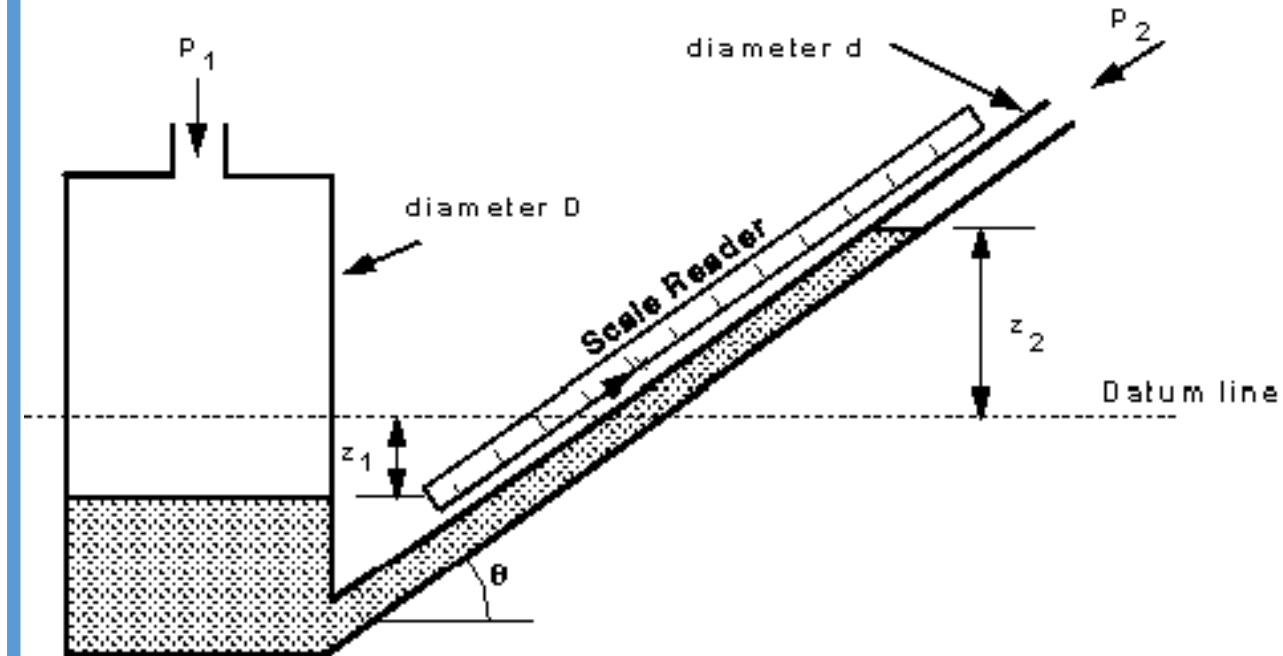
Inclined-tube manometers can be used to measure small pressure of gases accurately



$$p_A + \gamma_1 h_1 = p_B + \gamma_3 h_3 + \gamma_2 (l_2 \sin \theta)$$



$$p_A - p_B = \gamma_2 (l_2 \sin \theta) + \gamma_3 h_3 - \gamma_1 h_1$$



$$\text{Displaced volume} = A \times y = a \times s$$

where A and a are cross-sectional areas (C.S.A)



$$p_{\text{gas (abs)}} = \gamma [y + s \times \sin \theta] + p_{\text{atm}}$$

$$p_{\text{gas (gauge)}} = \gamma [y + s \times \sin \theta]$$

Variation of Pressure in Gases

Example :

In the Standard Atmosphere the temperature decreases linearly with altitude up to an elevation of 11.0 km. For a linear temperature variation with altitude given by $T = T_o - mz$, where m is constant. Find the pressure variation as a function of temperature.

$$dp = -\rho g dz = -\frac{\rho g}{RT} dz = -\frac{\rho g}{R(T_o - mz)} dz$$

$$\int_{p_o}^p \frac{dp}{p} = - \int_0^z \frac{g dz}{R(T_o - mz)}$$

$$\ln \left(\frac{p}{p_o} \right) = \frac{g}{mR} [\ln(T_o - mz)] \Big|_0^z = \frac{g}{mR} \ln \left(\frac{T_o - mz}{T_o} \right) = \frac{g}{mR} \ln \left(\frac{T}{T_o} \right)$$


$$p = p_o \left(\frac{T}{T_o} \right)^{\left(\frac{g}{mR} \right)}$$

Note:

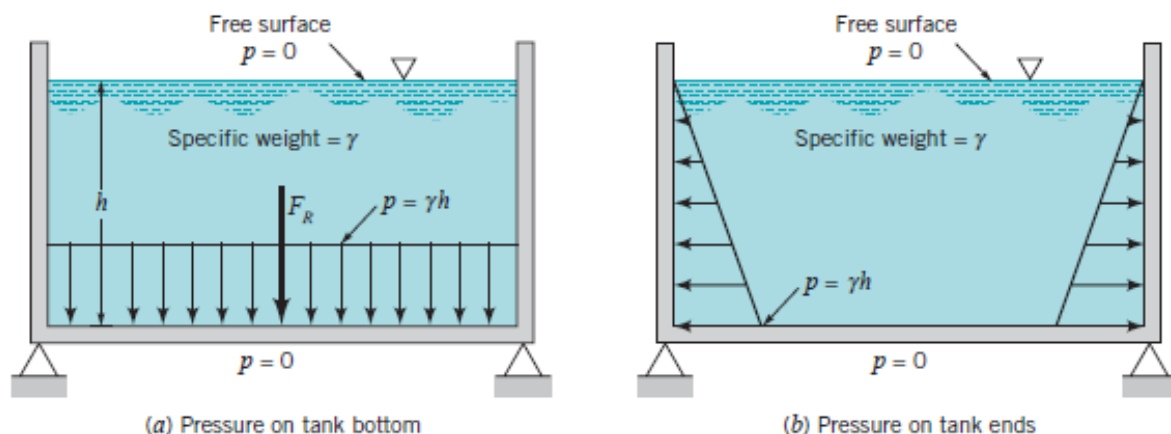
Variation of pressure in gases only counts if the height of the gas (h) is enough to generate a significant pressure difference (Δp).

Practical Application - Hydrostatic Force on Submerged Surfaces

When a surface is submerged in a fluid, forces develop on the surface due to the fluid.

The determination of these forces is important in the design of storage tanks, ships, dams, and other hydraulic structures

For fluids at rest we know that the force must be perpendicular to the surface since there are no shearing stresses present



■ **FIGURE 2.16** (a) Pressure distribution and resultant hydrostatic force on the bottom of an open tank. (b) Pressure distribution on the ends of an open tank.

Hydrostatic Force on Submerged Surfaces

An arbitrary plane surface of area A inclined at an angle θ below the free surface

$$\sin\theta = \frac{h}{y} = \frac{h_c}{y_c} \quad \Rightarrow \quad h = y \sin\theta$$

At any given depth, h , the force, dF acting on strip dA (the differential area) is:

$$dF = p \, dA = \gamma h \, dA = \gamma (y \sin\theta) dA$$

dF is perpendicular to the surface

Integrating over the plate area:

$$F_R = \int_A p \, dA = \int_A \gamma h \, dA = \int_A \gamma y \sin\theta \, dA$$

$$F_R = \gamma \sin\theta \int_A y \, dA$$

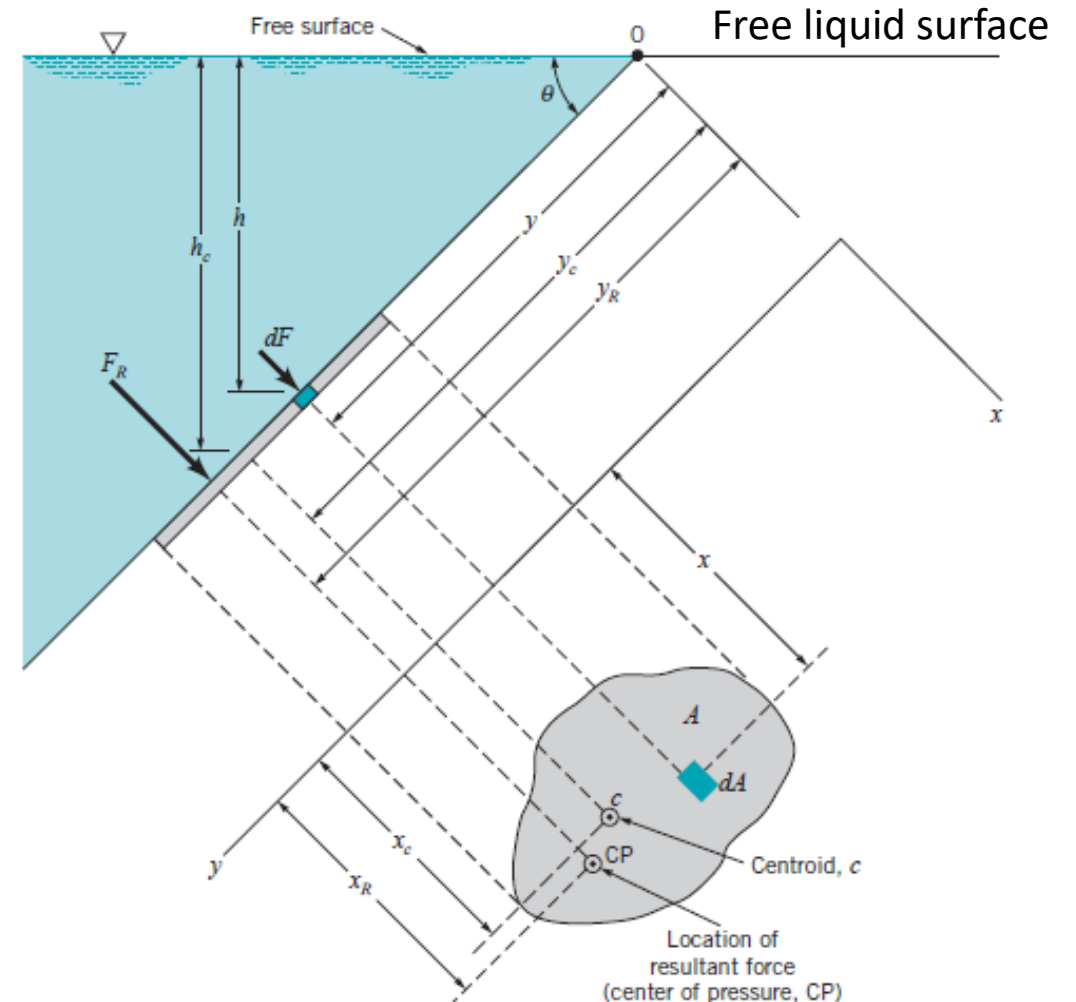
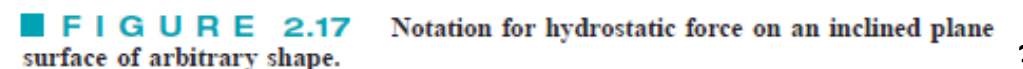


FIGURE 2.17 Notation for hydrostatic force on an inclined plane surface of arbitrary shape.

First moment of the area:

$$\int_{area} y \, dA = A \, y_c$$
$$F_R = \gamma A h_c$$

Note that the magnitude of the force is independent of the angle θ and depends only on the **specific weight of the fluid (γ)**, the **total immersed area (A)**, and the **depth of the centroid of the immersed area below the surface (h_c)**.



Hydrostatic Force on Submerged Surfaces

To find point of action of the resultant fluid force (center of pressure, C.P. or P) :

Sum of moments of infinitesimal forces, dF about OX
= Moment of the resultant fluid force, F about OX

$$F_R y_R = \int_A y dF$$

$$(\gamma A y_c \sin \theta) y_R = \int_A y (\gamma h dA)$$

$$(\gamma A y_c \sin \theta) y_R = \int_A \gamma \sin \theta y^2 dA = \gamma \sin \theta \int_A y^2 dA$$

$$y_R = \frac{\int_A y^2 dA}{y_c A}$$

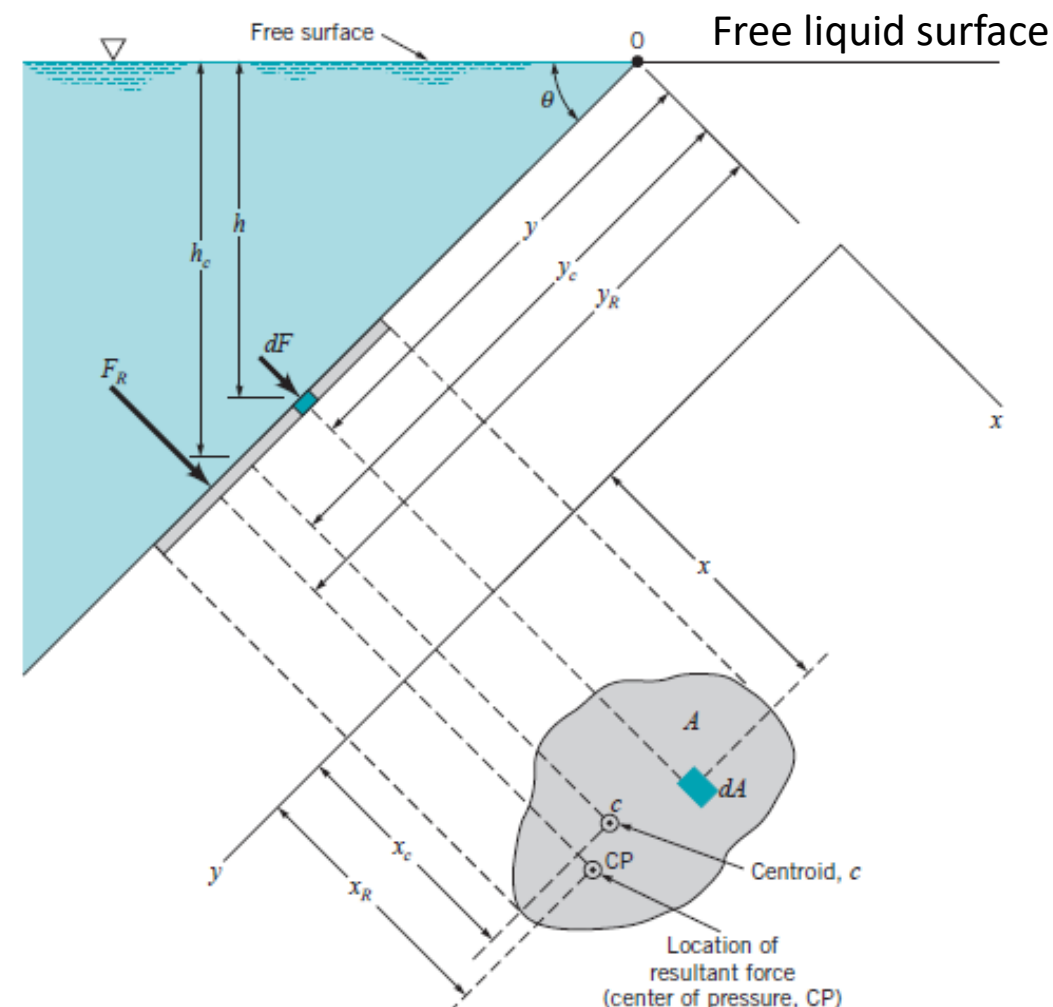


FIGURE 2.17 Notation for hydrostatic force on an inclined plane surface of arbitrary shape.

Hydrostatic Force on Submerged Surfaces

Second moment of the area (Moment of Inertia):

$$\int_A y^2 dA = I_x \quad \Rightarrow \quad y_R = y_{cp} = \frac{I_x}{y_c A}$$

The integral in the numerator is the **second moment of the area (moment of inertia)**, I_x , with respect to an axis formed by the intersection of the plane containing the surface and the free surface (x axis).

Use can now be made of the parallel axis theorem to express I_x :

$$I_x = I_{xc} + A y_c^2$$

$$y_R = y_{cp} = y_c + \frac{I_{xc}}{y_c A}$$

$$\Delta = y_P - y_C = \frac{I_C}{A y_C}$$

Similarly:

$$x_R = x_{cp} = x_c + \frac{I_{xyc}}{y_c A}$$

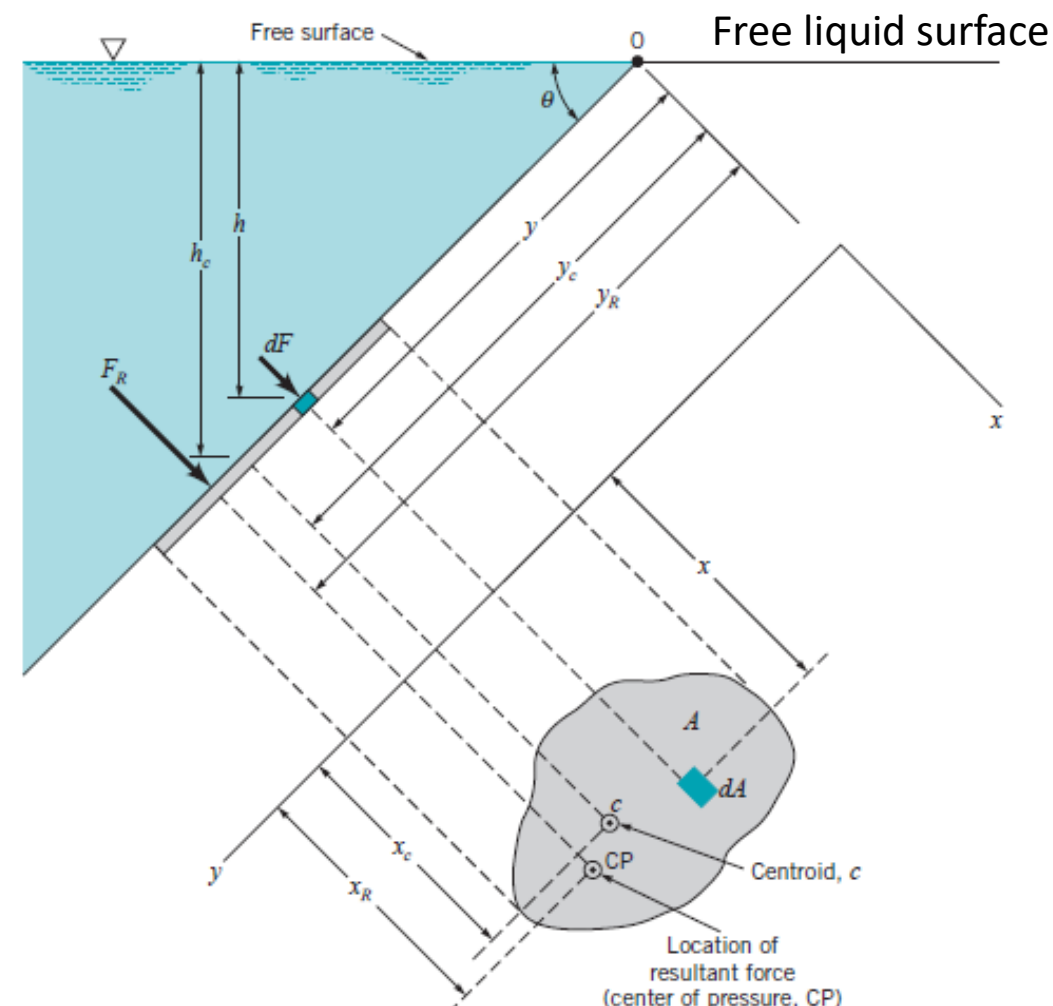
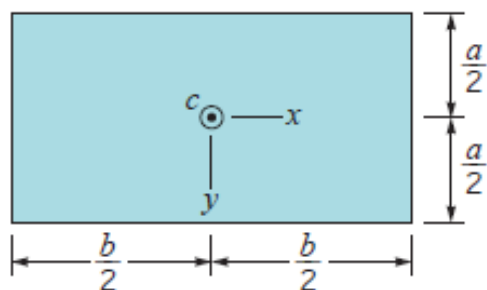


FIGURE 2.17 Notation for hydrostatic force on an inclined plane surface of arbitrary shape.

Centroidal Moments of Inertia for Various Cross Sections



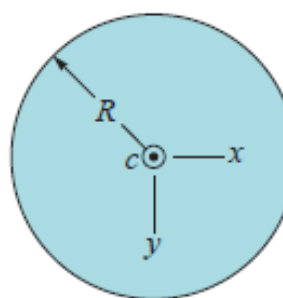
(a) Rectangle

$$A = ba$$

$$I_{xc} = \frac{1}{12} ba^3$$

$$I_{yc} = \frac{1}{12} ab^3$$

$$I_{xyc} = 0$$

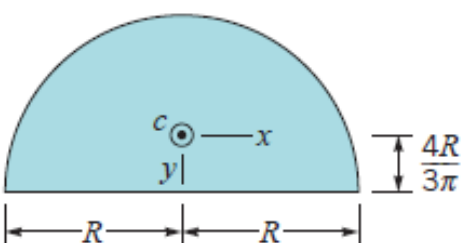


(b) Circle

$$A = \pi R^2$$

$$I_{xc} = I_{yc} = \frac{\pi R^4}{4}$$

$$I_{xyc} = 0$$



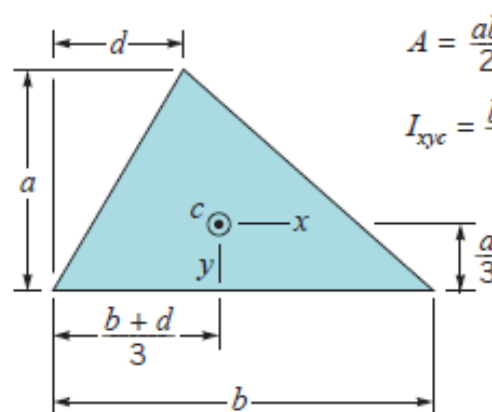
(c) Semicircle

$$A = \frac{\pi R^2}{2}$$

$$I_{xc} = 0.1098R^4$$

$$I_{yc} = 0.3927R^4$$

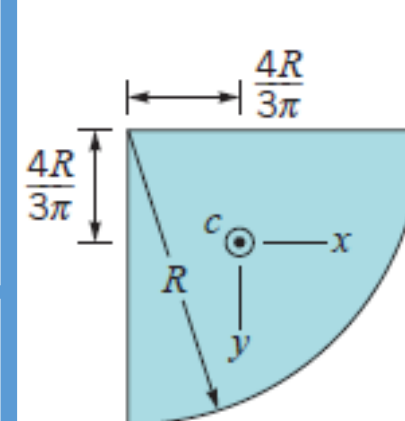
$$I_{xyc} = 0$$



(d) Triangle

$$A = \frac{ab}{2} \quad I_{xc} = \frac{ba^3}{36}$$

$$I_{xyc} = \frac{ba^2}{72}(b - 2d)$$



(e) Quarter circle

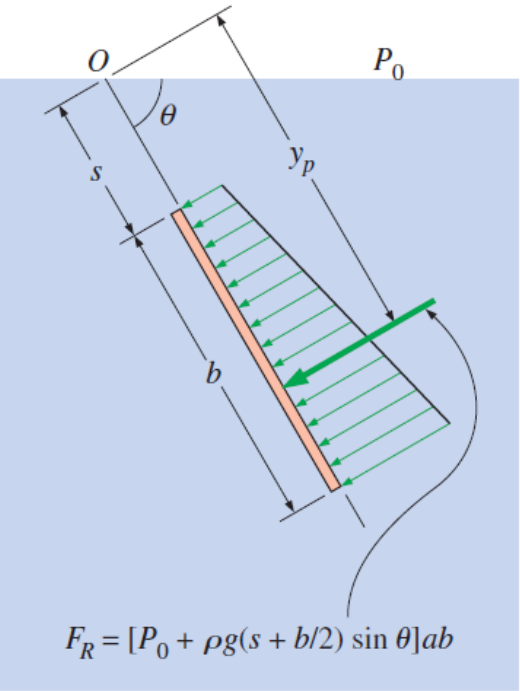
$$A = \frac{\pi R^2}{4}$$

$$I_{xc} = I_{yc} = 0.05488R^4$$

$$I_{xyc} = -0.01647R^4$$

Special Cases - Hydrostatic Force on Submerged Surfaces

Inclined (Tilted) Plate



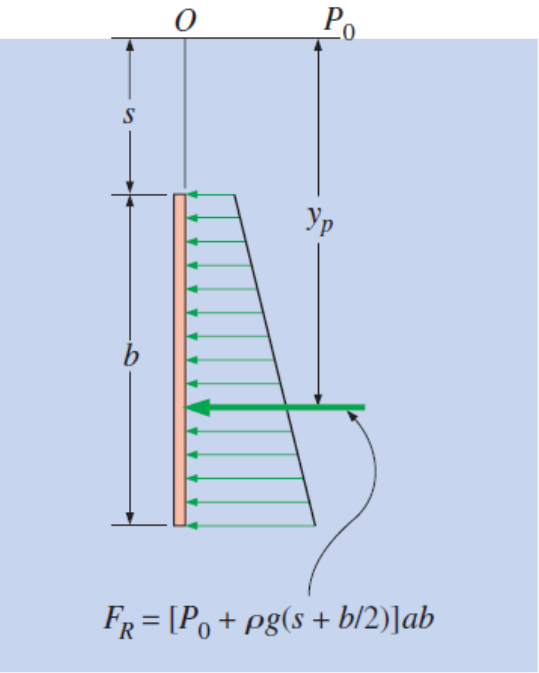
(a) Tilted plate

$F_R = \gamma A h_c$

$h_c = y_c \sin \theta$

$\Delta = y_P - y_C = \frac{I_C}{A y_C}$

Vertical Plane Surface



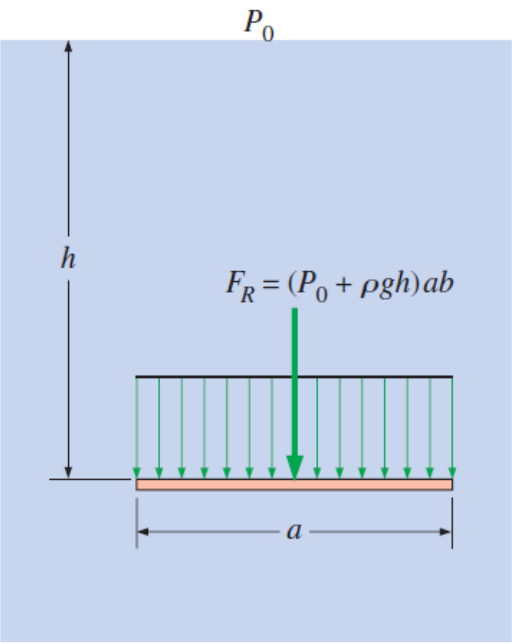
(b) Vertical plate

$\theta = 90^\circ \rightarrow \sin \theta = 1 \rightarrow y_c = h_c$

$F_R = \gamma A h_c$

$\Delta = y_P - y_C = \frac{I_C}{A y_C}$

Horizontal Plane Surface



(c) Horizontal plate

$F_R = \gamma A h_c$

C. P. $\equiv$ C. G.

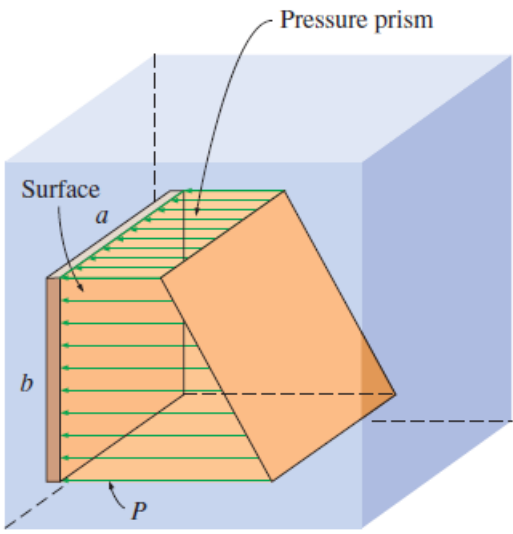


FIGURE 3-32
The hydrostatic forces acting on a plane surface form a pressure prism whose base (left face) is the surface and whose length is the pressure.

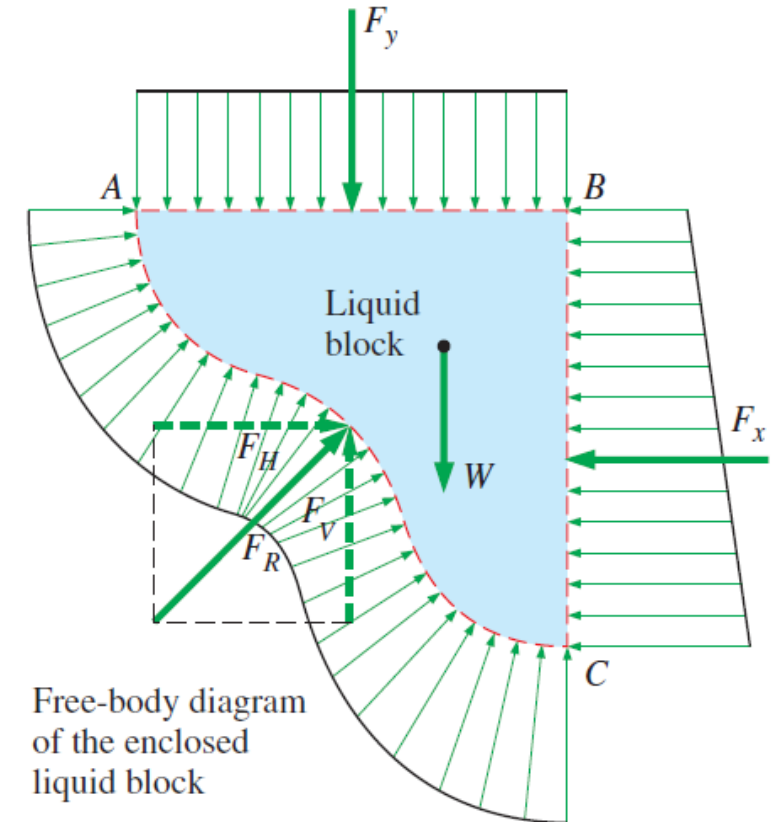
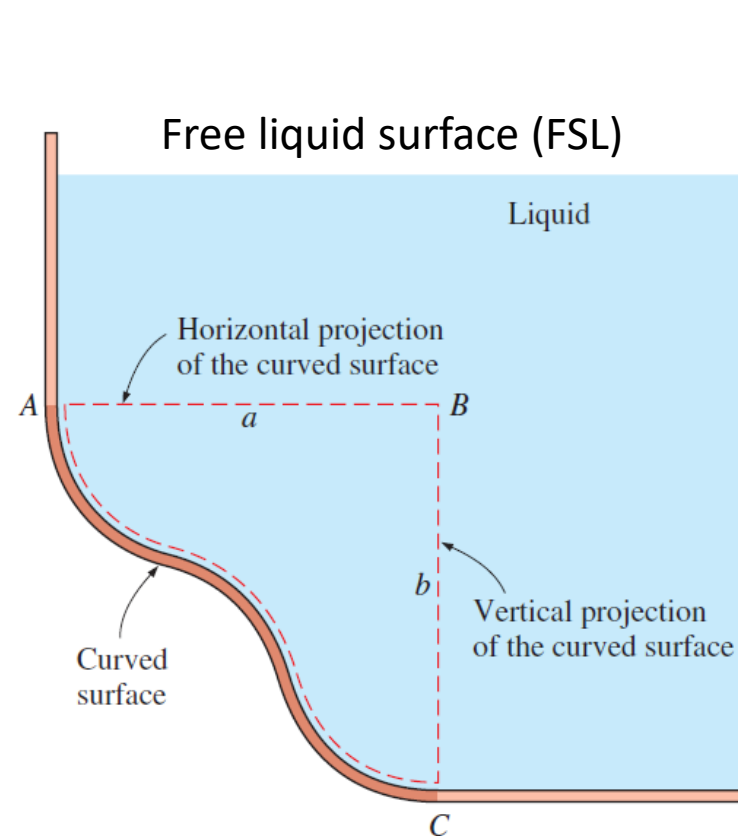
Hydrostatic Forces on a Curved Surface

Resultant fluid force, F , can not be obtained directly, since infinitesimal forces dF on each strip are in different directions (normal to the surface)

Instead, we calculate its components (horizontal F_x and vertical F_y)

$$F_{\text{Resultant}} = \sqrt{F_x^2 + F_y^2}$$

$$\tan\theta = \frac{F_y}{F_x}$$



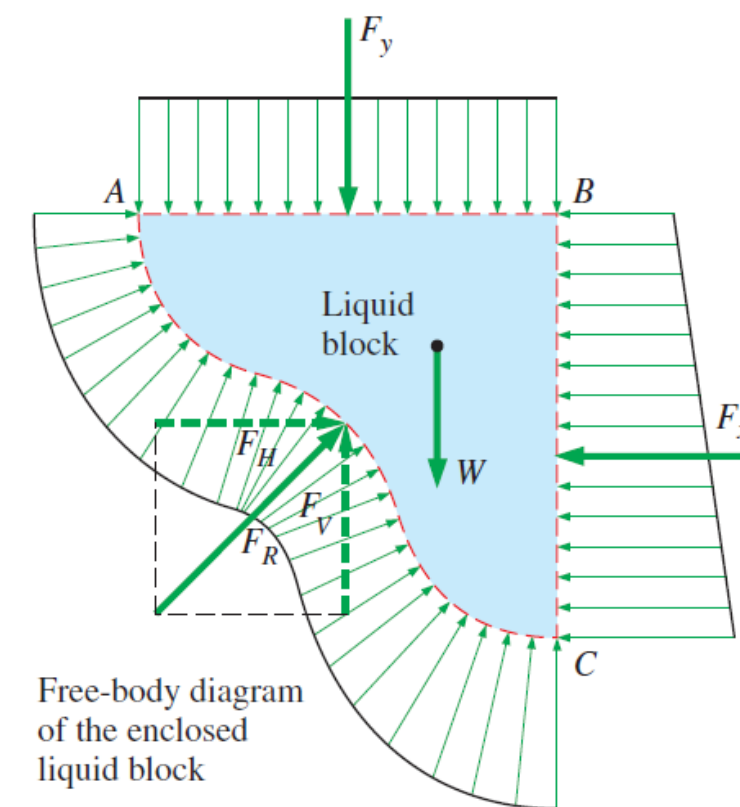
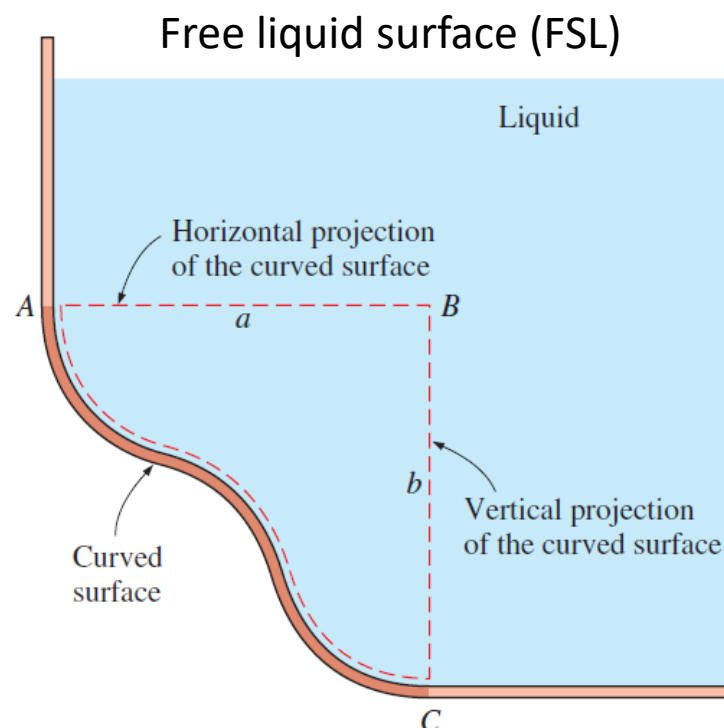
Hydrostatic Forces on a Curved Surface

F_x = Horizontal component of hydrostatic force on curved surface

Calculate F_x by calculate the force on the vertical projection of the curved surface (on the projection of curved surface on a vertical plan)

➔ $F_x = (\gamma A h_c)_{vertical\ proj}$

$$\Delta = y_P - y_C = \left(\frac{I_C}{A y_C} \right)_{vertical\ poj}$$



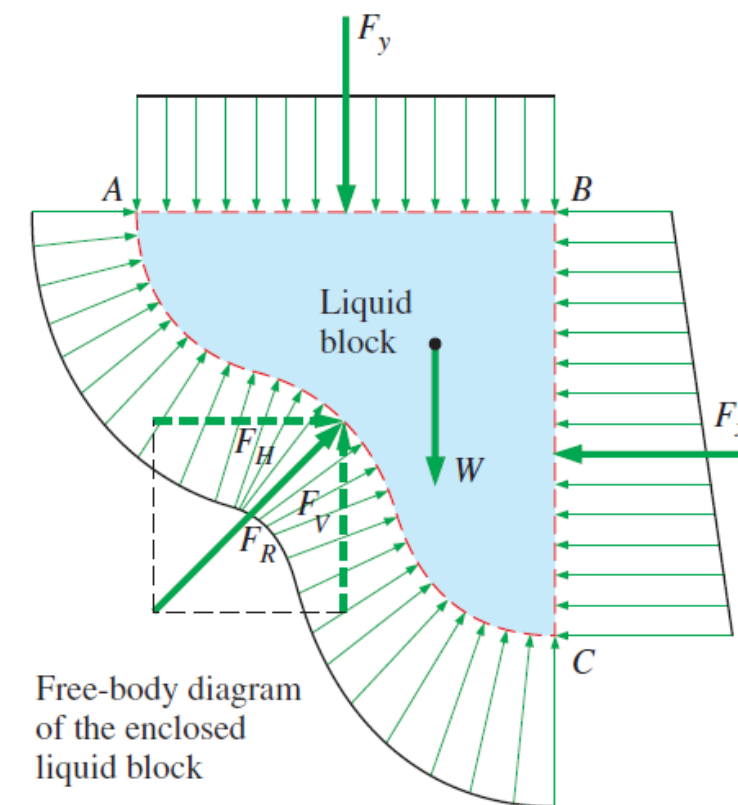
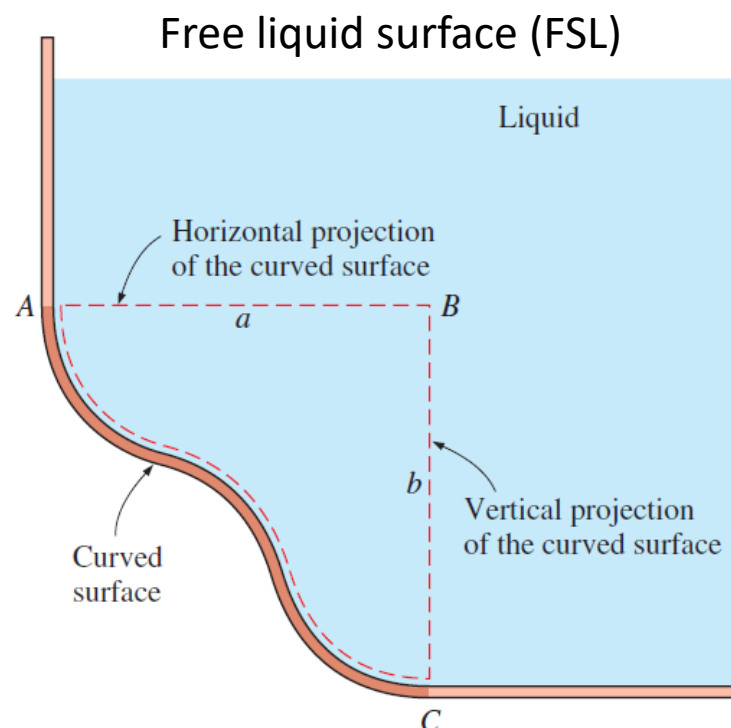
Hydrostatic Forces on a Curved Surface

F_y = Vertical component of hydrostatic force on curved surface

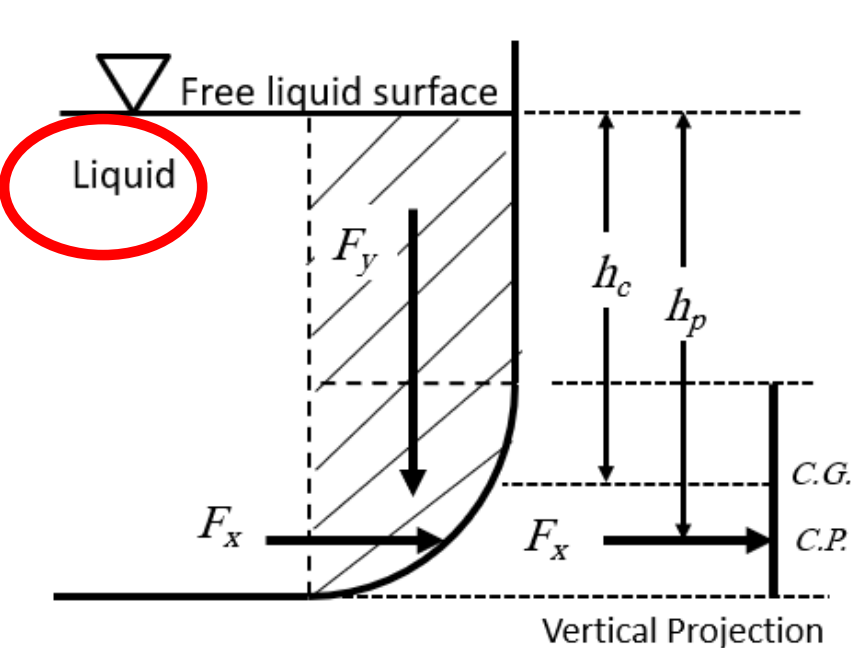
F_y = is the weight of actual liquid (or imaginary liquid) extending (confined) between the curved surface and the free liquid surface (FSL), and it acts through the centroid of this volume of the liquid

➔ $F_y = \gamma \nabla$

where ∇ is the volume of actual liquid (or imaginary liquid) extending (confined) between the curved surface and the free liquid surface (FSL)

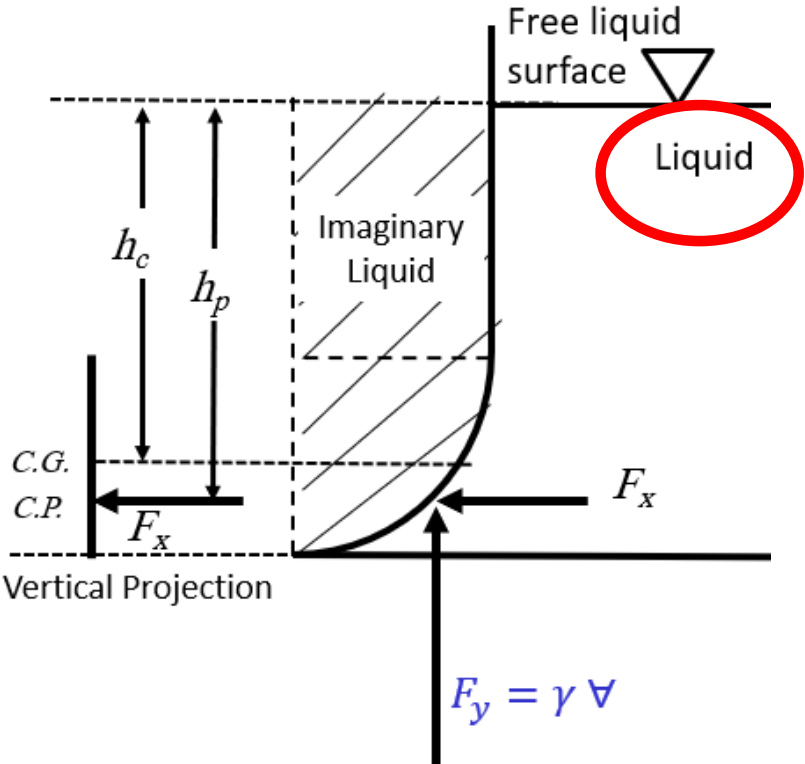


Special Cases: Hydrostatic Forces on a Curved Surface



F_y = weight of **actual (real)** liquid and acts downwards at the centroid of the hatched volume

↓ $F_y = \gamma \nabla$



F_y = weight of **imaginary** liquid and acts upwards at the centroid of the hatched volume

↑ $F_y = \gamma \nabla$

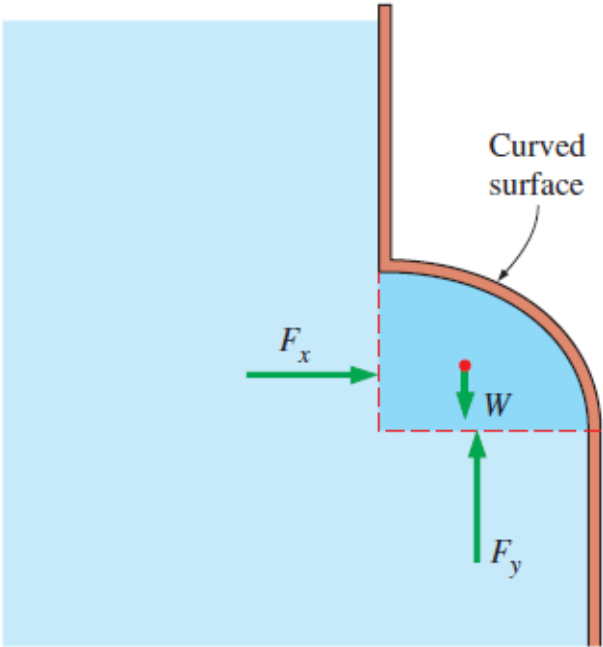


FIGURE 3-37
When a curved surface is above the liquid, the weight of the liquid and the vertical component of the hydrostatic force act in the opposite directions.